



NAPOLI LIBERAZIONE DIGITAL PLATFORM IMPLEMENTATION FOR GRAF S.R.L

Facoltà di Ingegneria dell'informazione, Informatica e Statistica

Corso di Laurea Magistrale in Ingegneria Gestionale

Student 1 - Mirco Soriano

Matr. 2205397

Team

400

Student 2 - Hanna Vollmo Oster

Matr. 2118620

Student 3 - Johan Sjolín

Matr. 2286061

Student 4 - Bogdan Andrei Tutuianu

Matr. 2217481

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ABSTRACT

The project analyzed in this report concerns the design and development of the digital platform “*Napoli Liberazione*”, aimed at preserving and enhancing the historical memory of the Four Days of Naples through innovative technological solutions. The initiative is promoted within a broader context of cultural valorization and digital transformation, with the objective of making historical and multimedia content accessible to a wide and global audience.

The project involves the creation of a web-based platform and mobile applications integrating advanced functionalities, including artificial intelligence systems, interactive storytelling, and multimedia content management. The platform is designed to offer an engaging and educational user experience, enabling users to explore historical documents, audiovisual materials, and digital archives in an intuitive and accessible way.

From a project management perspective, this initiative represents a complex and interdisciplinary endeavor, combining technological development, cultural dissemination, and user-centered design. The project aims not only to digitize historical heritage but also to foster social engagement and cultural awareness, ensuring accessibility for users with different needs and levels of digital literacy.

The report will analyze the methodologies, tools, and management approaches adopted throughout the project lifecycle, evaluating the effectiveness of planning, execution, and control activities. Furthermore, particular attention will be given to the identification of critical issues and potential areas for improvement, in order to derive useful insights and recommendations for enhancing similar projects in the future. Finally, a critical assessment of the results achieved will be provided, highlighting key success factors, challenges encountered, and lessons learned.

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EXECUTIVE SUMMARY

This report presents the execution and evaluation of the "Napoli Liberazione" project, carried out by Partenope Project Lab and commissioned by Graf S.R.L.. The initiative aimed to design and develop a digital platform to preserve and enhance the historical memory of the Four Days of Naples through innovative technological solutions.

In the first chapter, we introduce the overall goals of the project and explain its specific objectives within the broader context of cultural valorization and digital transformation. We provide background information about both the client, Graf S.R.L., and the developing company, Partenope Project Lab, and reconstruct the original business case and project charter that justified the initiative and its alignment with the TOCC public funding program.

In the second chapter, we describe how the project scope was defined and planned. We go through the technical and functional requirements—such as multimedia storytelling, AI integrations, and accessibility features—the high-level and detailed planning structured via a Work Breakdown Structure, and the human resources, procurement, and risk management strategies adopted to guide the project toward successful execution.

The third chapter focuses on the execution and control phase. Here, we analyze how the project was managed in practice, discussing scope monitoring, quality assurance through continuous testing, progress tracking via periodic SAL meetings, and strict cost management. We also report on unforeseen events, specifically the severe delays and poor-quality deliverables from an externally hired Senior Developer, and how the team adapted through rapid resource reallocation and crashing techniques to protect the schedule and budget.

Finally, in the fourth chapter, we reflect on the closure of the project and conduct a critical evaluation. We summarize the results achieved—delivering the full technical scope on time and within the €57,000 budget constraint—highlight the key success factors like internal flexibility and transparent communication, and draw lessons learned regarding supplier management and contingency planning that could be applied to future digital initiatives.

CHAPTER 1.

PROJECT GOALS AND CONTEXT

1.1. PROJECT DESCRIPTION

The project “Napoli Liberazione” consists of the design and development of a customized software platform commissioned by Graf S.R.L. and developed by Partenope Project Lab. The initiative is positioned within a cultural and artistic context, with the aim of supporting the dissemination and enhancement of the historical memory related to the Four Days of Naples through digital technologies.

The Four Days of Naples refer to a popular uprising that took place in September 1943, during World War II, when the citizens of Naples revolted against the German occupation, leading to the liberation of the city before the arrival of Allied forces. This historical event represents a significant example of collective resistance and civic engagement, forming the cultural foundation of the project.

The core output of the project is a digital platform, including web-based solutions and related applications, designed to manage and distribute multimedia content through a streaming-oriented architecture. The platform enables the organization, storage, and accessibility of historical and cultural materials, providing users with an integrated and interactive digital experience.

From a functional perspective, the system is characterized by the integration of multimedia storytelling, allowing users to explore historical events through audiovisual narratives, a structured digital archive for the preservation and retrieval of documents and content, and advanced accessibility features aimed at ensuring inclusiveness for users with different needs.

The platform is primarily targeted at citizens, students, and researchers, offering them a tool to access, explore, and interact with cultural heritage in an intuitive and engaging way. In this sense, the project not only represents a technological development initiative but also a strategic effort to promote cultural awareness and social engagement through digital innovation.

Overall, the project can be classified as a complex and interdisciplinary endeavor, involving the coordination of technical, cultural, and organizational components, and requiring a structured project management approach to ensure the achievement of its objectives within the defined constraints.

1.2. COMPANY DESCRIPTION

Partenope Project Lab is a strategic consulting and integrated design company that operates in support of enterprises, public institutions, and third-sector organizations. The company is characterized by a

multidisciplinary approach, combining economic, technological, and cultural competencies to foster sustainable development and innovation-oriented projects.

The core mission of Partenope Project Lab is to support the growth of innovative initiatives, facilitate access to public funding and European programs, and design solutions that generate both economic and social value. In this context, the company frequently collaborates with clients applying for public funding opportunities, providing both technical and legal support in the preparation of project proposals and applications for national and regional calls, such as programs related to territorial development and cultural regeneration.

Its operational model is based on a structured process that includes needs analysis, strategic planning, technical and financial design, and continuous support during the implementation and evaluation phases. This approach allows the company to guide clients throughout the entire project lifecycle, from the initial concept to the final reporting and impact assessment.

The company operates across multiple domains, including business consulting, technological development, and cultural project management. In particular, it provides services related to software development, digital transformation, and the organization of cultural and artistic initiatives, demonstrating a strong capability to integrate technology with content creation and dissemination.

A key strength of Partenope Project Lab lies in its ability to build synergies between public and private actors, fostering collaborative networks and supporting projects with a strong territorial and social impact. This positioning makes the company particularly suitable for initiatives such as “Napoli Liberazione,” where technological innovation is combined with cultural valorization and community engagement.

Overall, Partenope Project Lab can be considered a flexible and innovation-driven organization, capable of managing complex and interdisciplinary projects through a systemic and impact-oriented approach.

1.3. BUSINESS CASE

BACKGROUND

The “Napoli Liberazione” project was commissioned by Graf S.R.L. and developed by Partenope Project Lab within the broader context of cultural heritage valorization and digital transformation initiatives. The project originates from the need to preserve and enhance the historical memory of the Four Days of Naples, a key event in Italian history, by leveraging modern digital technologies.

The initial context highlighted a fragmentation and limited accessibility of historical content, often dispersed across different sources and not easily available to the general public, especially younger generations. This situation created the necessity for a centralized and accessible digital solution capable of integrating historical documents, multimedia content, and direct testimonies into a unified platform.

In response to these challenges, the project aims to provide an innovative and inclusive digital environment that facilitates the dissemination of cultural heritage and encourages active user engagement. Furthermore, the initiative is aligned with public funding opportunities, such as the TOCC program, which supports digital and cultural projects aimed at promoting innovation and territorial development.

BUSINESS OBJECTIVE

The primary business objective of this initiative is to provide the client with an effective digital solution to enhance the dissemination of cultural heritage and historical knowledge. In particular, the goal is to promote cultural content related to the Four Days of Naples, increasing its visibility and accessibility to a broader and more diverse audience.

Furthermore, the initiative aims to generate added value for the client's cultural activities by strengthening its positioning within the cultural and educational landscape. Through the development of a dedicated digital platform, the solution enables the expansion of communication channels, facilitates user engagement, and improves the overall accessibility of content.

Overall, the project supports the client's strategic objective of leveraging digital innovation to foster cultural awareness, enhance public engagement, and create long-term value for the organization and its stakeholders.

PROJECT SCOPE

The scope of the project includes the design and development of a web-based platform aimed at the dissemination of historical and cultural content related to the Four Days of Naples. The platform is intended to serve as a centralized digital environment where users can access and explore different types of multimedia materials.

Specifically, the project encompasses the creation of a website that integrates narrative content, images, and video materials, providing a structured and engaging representation of the events of the Neapolitan resistance and its key historical figures. The platform is designed to support the organization, storage, and visualization of such content in an accessible and user-friendly manner.

The scope also includes the development of functionalities that enable users to navigate and interact with the available materials, ensuring a coherent and intuitive user experience. At a high level, the project focuses exclusively on the digital platform and its content management capabilities, without including additional physical installations or offline services.

CURRENT SITUATION AND PROBLEM - OPPORTUNITY STATEMENT

The current situation highlights significant limitations in the accessibility and preservation of cultural content related to the historical memory of the Four Days of Naples. Much of this knowledge is traditionally transmitted through informal channels, such as oral storytelling across generations, which exposes it to the risk of being progressively lost over time if not properly documented and preserved.

In addition, cultural associations involved in the promotion of such initiatives often face structural challenges in achieving economic sustainability. These organizations typically operate with limited financial resources and encounter difficulties in generating stable revenue streams, which constrains their ability to invest in long-term cultural projects and digital innovation.

However, this context also presents a relevant opportunity. The increasing availability of public funding programs, such as the TOCC initiative, creates favorable conditions for the development of digital solutions aimed at cultural preservation and dissemination. By leveraging these funding opportunities, it becomes possible to transform existing limitations into a structured and sustainable project capable of enhancing both cultural impact and organizational resilience.

CRITICAL ASSUMPTION AND CONSTRAINTS

The development of the project is based on several key assumptions. Firstly, it is assumed that there is a sufficient level of interest from potential users, including citizens, students, and researchers, in accessing and engaging with historical and cultural content related to the Four Days of Naples. Secondly, it is assumed that relevant multimedia materials, such as images, videos, and audio content, are available and can be effectively collected and integrated into the platform.

At the same time, the project is subject to a number of critical constraints. One of the main constraints is the limited timeframe, which is strongly influenced by the requirements and deadlines associated with public funding programs. Additionally, the project must be developed within a limited budget, requiring careful allocation and management of financial resources.

Finally, organizational constraints must also be considered. The development is carried out by a relatively small company managing multiple projects simultaneously, which may limit the availability of technical and human resources and impact the overall execution capacity of the initiative.

ANALYSIS OF OPTIONS AND RECOMMENDATION

Several alternative solutions were considered in order to address the need for preserving and disseminating the historical memory of the Four Days of Naples.

The first option consisted of maintaining a traditional physical archive, based on the collection and preservation of historical materials in non-digital formats. While this solution ensures the conservation of original documents, it significantly limits accessibility and does not support wide dissemination, especially for younger and geographically distant audiences. However, this option was not considered viable, as the project is developed within the framework of a public funding program specifically aimed at digitalization initiatives. As a result, non-digital solutions were excluded due to misalignment with the objectives and requirements of the funding scheme.

The second option involved the development of a basic static website, providing textual information and limited multimedia content. Although this approach would improve accessibility compared to traditional methods, it would not fully exploit the potential of digital technologies in terms of user engagement, interactivity, and scalability.

The third option, which has been selected as the preferred solution, consists of the development of a comprehensive digital platform integrating multimedia storytelling, structured content management, and accessibility features. This solution provides a more effective and scalable approach, enabling a richer user experience and ensuring broader dissemination of cultural content.

Based on this analysis, the implementation of an advanced digital platform is recommended, as it represents the only solution fully aligned with the project objectives, funding requirements, and long-term sustainability goals.

PRELIMINARY PROJECT REQUIREMENTS

The preliminary project requirements define the main functionalities that the platform must provide in order to achieve its objectives.

First, the system must enable users to search and access relevant content efficiently, supported by artificial intelligence mechanisms that personalize the user experience based on individual preferences. Additionally, the platform should offer an interactive digital environment where users can navigate multimedia content, including videos with multi-scene storytelling and related textual or audiovisual insights accessible through links or QR codes.

Furthermore, the platform must support user interaction and engagement through features such as user registration, access to reserved areas with exclusive content, and the possibility to provide feedback and comments. It should also allow users to subscribe to a newsletter and share or download selected multimedia materials in order to promote cultural dissemination.

Another key requirement is the support of multilingual access, ensuring that content can be accessed by a broader audience. Finally, the system must guarantee a high level of accessibility, including features such as screen readers, subtitles, audio descriptions, and other tools designed to support users with disabilities.

Overall, these requirements define a user-centered, inclusive, and technologically advanced platform aimed at maximizing accessibility, engagement, and cultural impact.

BUDGET ESTIMATE

The preliminary budget for the development of the project is estimated at approximately €57,000. This budget reflects the overall effort required to design, develop, and implement the digital platform, including software development, content integration, and testing activities.

The allocation of financial resources is expected to cover the main phases of the project lifecycle, such as requirements analysis, system design, development, deployment, and initial maintenance. Given the limited financial availability, careful cost management will be essential to ensure that the project is completed within the defined budget while maintaining the required quality standards.

The budget is also aligned with the funding framework of public programs, such as the TOCC initiative, which supports digital and cultural projects. Therefore, the financial plan must comply with the constraints and reporting requirements associated with such funding schemes.

SCHEDULE ESTIMATE AND MILESTONE CHART

The overall duration of the project is approximately seven months. This timeframe reflects the complexity of the initiative and the need to complete all phases of the software development lifecycle, from initial analysis to final delivery and validation.

The project is structured around a series of key milestones that guide its progression. The first milestone is represented by the formal signing of the contract, which marks the official start of the project. This is followed by the execution of the functional analysis phase, during which customer journeys and system requirements are defined.

Subsequently, an initial release of the platform is delivered in a testing environment, allowing for user validation and the identification of potential issues. Based on the feedback collected, a refined version of the platform is developed and delivered for a second phase of user testing, during which defects are identified and corrected.

The final milestone consists of the delivery of the definitive version of the platform, following the successful completion of user acceptance tests and the resolution of all identified issues. This structured sequence of milestones ensures a controlled and iterative development process, aimed at guaranteeing quality and alignment with user expectations.

BENEFITS STATEMENT

The project is expected to generate multiple benefits across cultural, social, and economic dimensions.

From a cultural perspective, the platform will contribute to the preservation and dissemination of historical knowledge related to the Four Days of Naples, ensuring that such heritage is accessible to a wider audience and safeguarded for future generations. From a social standpoint, the initiative promotes community engagement and cultural awareness, encouraging active participation from citizens, students, and researchers.

In addition to these aspects, the project is also expected to produce economic benefits. Indirectly, the platform may contribute to the promotion of the local territory and cultural tourism by increasing the visibility of historical sites and narratives associated with the city. Moreover, a key economic advantage lies in the ownership of the developed software, which represents a valuable digital asset for the promoting organization. This asset can be further leveraged for future initiatives, collaborations, or potential monetization opportunities, thus increasing both the direct and indirect value of the organization.

Overall, the combination of cultural impact, social engagement, and economic potential makes the project a sustainable and value-generating initiative.

POTENTIAL RISK AND RISK REGISTER

The project is exposed to several potential risks that may affect its successful completion, particularly in terms of schedule and quality.

One of the main risks is related to possible delays in project execution. Since the development activities are carried out by a relatively small organization managing multiple projects simultaneously, there is a risk that resource allocation constraints may lead to slower progress than initially planned.

Another critical risk concerns the involvement of third-party providers responsible for the development of specific components of the platform. The reliance on external contributors may introduce uncertainties in terms of quality and consistency of the delivered components. In particular, there is a risk that externally developed parts may not fully meet the required standards or may require additional effort for proper integration within the overall system.

These factors may lead to extended development times, increased coordination efforts, and potential delays in the final delivery. To mitigate these risks, careful planning, continuous monitoring, and effective communication between all involved parties will be essential throughout the project lifecycle.

GOVERNANCE CONSIDERATIONS

The governance structure of the project is based on a clear distribution of roles and responsibilities between the client and the development partner. The final decision-making authority is assigned to the client, Graf S.R.L., which defines the strategic direction of the initiative. Partenope Project Lab plays a key advisory role, providing technical and strategic support throughout the project lifecycle.

During the initial phase, several coordination meetings are conducted to define the project objectives, identify feasible technical solutions, and align expectations between stakeholders. These meetings are essential to ensure a shared understanding of requirements and priorities.

As the project progresses, governance is maintained through a milestone-based approach. Each milestone represents a control point where the work completed is reviewed and validated, ensuring alignment with the defined objectives and allowing for adjustments when necessary. This structured approach supports effective monitoring, facilitates communication among stakeholders, and reduces the risk of misalignment during the execution phase.

1.4. PROJECT CHARTER

Project Name : Napoli LiberAzione

Company Name : Parthenope Project Lab

Date : 03/2023

Project Customer : Graf S.R.L.

BACKGROUND

The project originates from the need to preserve and disseminate the historical memory of the Four Days of Naples through modern digital technologies. Historical and cultural content related to this event is often fragmented and not easily accessible, creating the risk of progressive loss of valuable testimonies and materials over time.

To address this issue, the initiative aims to develop a digital platform capable of collecting, organizing, and distributing multimedia content in an accessible and engaging way. The project combines cultural valorization with digital innovation, providing users with an interactive environment designed to enhance the accessibility and dissemination of historical knowledge.

PROJECT OBJECTIVES

The main objective of the project is to develop a digital platform aimed at preserving and disseminating the historical memory of the Four Days of Naples through accessible and innovative technological solutions.

The initiative also aims to enhance the accessibility of cultural content, improve user engagement through multimedia experiences, and provide a structured digital environment for the management and distribution of historical materials.

PROJECT SCOPE

The scope of the project includes the design and development of a web-based platform dedicated to the preservation and dissemination of historical and cultural content related to the Four Days of Naples.

The platform will provide users with access to multimedia materials, including photographs, videos, and narrative content, through an interactive and accessible digital environment designed to support cultural engagement and knowledge sharing.

NECESSITY AND BENEFITS OF BUSINESS

The project is necessary to ensure the preservation and accessibility of historical and cultural content that could otherwise become progressively inaccessible or lost over time. Through digital technologies, the initiative supports the dissemination of historical knowledge and promotes greater cultural awareness among different categories of users.

The project is expected to generate both social and economic benefits. From a cultural and social perspective, it enhances public engagement and facilitates access to educational and historical materials. From an economic perspective, the platform represents a valuable digital asset for the promoting organization, contributing to its visibility, long-term value creation, and potential future development opportunities.

IN-SCOPE

- Design and development of the web platform dedicated to the dissemination of historical and cultural content related to the Four Days of Naples.
- Integration and management of multimedia materials, including photographs, videos, podcasts, and narrative content.
- Implementation of accessibility features aimed at improving usability for users with disabilities.
- Development of content management functionalities and user interaction features.

- Creation of a structured and accessible digital environment for cultural engagement and knowledge sharing.

OUT-OF-SCOPE

- Development of dedicated mobile applications.
- Management or supply of hardware infrastructures and physical devices.
- Long-term maintenance and operational support activities after the final project delivery.
- Physical preservation or restoration of historical documents and materials.
- Monetization systems, e-commerce functionalities, or commercial transactions through the platform.

FLEXIBILITY MATRIX

FLEXIBILITY MATRIX		
CONSTRAINT	FLEXIBILITY LEVEL	DESCRIPTION
 TIME	LOW	Project deadlines are strict due to funding requirements and contractual obligations. There is limited flexibility in adjusting the schedule.
 SCOPE	LOW	The project scope is clearly defined and includes specific deliverables and functionalities. Changes to scope are limited.
 COST	MEDIUM	The budget is limited but some flexibility is possible to accommodate necessary adjustments during development.
 QUALITY	MEDIUM	Quality standards are important, but minor adjustments can be considered to ensure project completion and stakeholder satisfaction.

Figure 1: Constraints and Flexibility

SCHEDULE OF OBJECTIVES AND DELIVERED

The project schedule is structured around a sequence of key objectives and deliverables designed to ensure progressive validation and controlled project execution.

Milestone	Deliverable
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Contract Signature	Official project approval and initiation
Functional Analysis Completion	Delivery and validation of the functional analysis document by the client
Test Release Delivery	Delivery of the platform release in a testing environment for user validation
Pre-Final Release Delivery	Delivery of the refined platform version for final testing and correction of identified defects
Final Release Delivery	Delivery of the definitive validated platform after successful user acceptance testing

Table 1: Milestones and Deliverables

SUCCESS CRITERIA

The project will be considered successful if the platform is fully functional and capable of delivering the expected multimedia and cultural content in a stable and accessible manner.

Another important success criterion is the completion of the project within the planned schedule, or with only limited and manageable delays. In addition, the final release of the platform should not present critical software defects that could compromise usability or system performance.

The project success will also depend on the positive evaluation and validation provided by the tutors and reviewers responsible for monitoring the initiative within the framework of the funding program. Overall, the achievement of technical reliability, schedule compliance, and stakeholder satisfaction will represent the key indicators of project success.

HIGH-LEVEL REQUIREMENTS

The project requires the development of a digital platform capable of supporting the management and dissemination of multimedia historical content related to the Four Days of Naples.

The platform must include accessibility features to ensure usability for different categories of users, including individuals with disabilities. In addition, the system should provide interactive storytelling functionalities aimed at enhancing user engagement through multimedia experiences.

Finally, the platform must support user registration functionalities and provide reserved areas for authenticated users, enabling access to dedicated content and interaction features.

RISKS

RISK ID	RISK DESCRIPTION	POTENTIAL IMPACT	LIKELIHOOD	RISK LEVEL
 R01	Development delays Delays in development activities due to limited internal resources and concurrent projects managed by the company.	 Schedule slippage and potential impact on milestones and final delivery.	Medium	MEDIUM
 R02	Integration issues with third-party components Difficulties in integrating code and components developed by external providers.	 Integration rework, increased testing effort and possible delays.	High	HIGH
 R03	Limited availability of resources Scarcity of key technical resources may affect development capacity.	 Reduced productivity and potential delays in critical activities.	Medium	MEDIUM
 R04	Quality inconsistencies in outsources components Variability in code quality from external suppliers may affect platform stability and performance.	 Increased defect rates, higher rework effort and potential impact on user experience.	Medium	MEDIUM
	RISK MANAGEMENT APPROACH Risks will be continuously monitored throughout the project lifecycle. Mitigation actions include proactive planning, regular progress tracking, and strong coordination with all stakeholders and external partners.			

Figure 2: Risks

STAKEHOLDERS

The project involves both core and non-core stakeholders, each characterized by different levels of influence and interest throughout the project lifecycle.

The core stakeholders include Graf S.R.L., acting as the client and decision-making authority, Partenope Project Lab, responsible for project coordination and development activities, and the technical development team involved in the implementation of the platform functionalities.

Non-core stakeholders include citizens, students, researchers, and cultural organizations interested in accessing and benefiting from the digital platform and its multimedia historical content. Although these stakeholders are not directly involved in strategic decision-making activities, they represent the primary users and beneficiaries of the project outcomes.

STAFFING DECISION

The project team is composed of a relatively small internal structure responsible for managing multiple projects simultaneously. For this reason, part of the technical development activities is supported by external developers specialized in software implementation.

The organizational structure is coordinated by the Project Manager, who supervises the overall execution of the project and ensures alignment between stakeholders, objectives, and deliverables. A dedicated legal team

is responsible for supporting the administrative and documentation activities related to participation in the funding initiative.

The software development team is responsible for the implementation of the platform functionalities. Within this team, the Software Team Leader coordinates the technical activities and supervises developers and testers involved in the development and validation processes.

TECHNICAL DECISION

All technical decisions related to the development of the platform are defined by the software development team according to the project requirements and implementation needs. These decisions are subsequently reviewed and confirmed by the Project Manager in order to ensure consistency with the project objectives, constraints, and overall strategic direction.

APPROVALS

The approval of the project and its main deliverables is primarily assigned to the client, which is responsible for validating the results produced throughout the project lifecycle. The Project Manager is responsible for supervising the operational progress of the project and validating the completion of activities and milestones before submission to the client.

In addition, the final outcomes of the project are subject to evaluation and acceptance by the tutors and reviewers responsible for monitoring the funding initiative. Their approval is necessary to confirm the compliance of the project with the objectives, requirements, and expected standards defined within the funding framework.

INITIAL COMMUNICATION PLAN

The communication management approach is structured to ensure continuous alignment between the internal project team and the client throughout the project lifecycle.

Within Partenope Project Lab, communication activities include weekly email updates regarding project progress, together with daily calls and meetings aimed at coordinating operational activities and ensuring alignment among team members.

Communication with the client is mainly organized around project milestones. At each milestone, formal email communications are used to notify the completion of activities and deliverables, followed by meetings or calls dedicated to presenting the work performed, discussing feedback, and validating project progress.

1.5. DETAILED DESCRIPTION OF PROJECT GOALS

1.5.1. PROJECT OUTPUT

The primary output of the project is the development and deployment of a web-based digital platform designed to preserve, organize, and disseminate historical and cultural content related to the Four Days of Naples. The platform aims to transform historical memory into an accessible and interactive digital experience capable of engaging different categories of users, including citizens, students, and researchers.

The platform integrates multiple types of multimedia content, such as photographs, videos, podcasts, articles, and narrative materials, allowing users to explore historical events through a dynamic and immersive approach. Particular attention is dedicated to storytelling functionalities, enabling users to navigate historical narratives through multimedia experiences and interconnected content.

In addition to content visualization, the system includes several user-oriented functionalities, such as content search tools, reserved areas for registered users, and interaction features designed to improve engagement and participation. Accessibility also represents a fundamental aspect of the platform, with the inclusion of functionalities aimed at supporting users with disabilities and ensuring a broader and more inclusive access to cultural content.

From a technical and operational perspective, the final project output corresponds to the definitive validated release of the platform, completed after testing, correction of identified defects, and approval by both the client and the tutors responsible for evaluating the project within the funding initiative. The resulting platform therefore represents not only a technological deliverable, but also a long-term digital asset capable of generating cultural, social, and organizational value over time.

1.5.2. PROJECT DURATION

The overall duration of the project is approximately seven months. The project timeline includes the execution of functional analysis activities, software development, testing phases, user validation, and final platform release.

The development process is organized through progressive milestones that allow continuous monitoring of project progress and validation of deliverables throughout the project lifecycle.

1.5.3. PROJECT BUDGET

The estimated project budget is approximately €57,000. The financial resources allocated to the project are mainly intended to support software development activities, testing, coordination, and content integration processes.

The budget structure is strongly influenced by the constraints and requirements associated with the funding initiative supporting the project. As a result, careful financial management and continuous monitoring of costs are essential to ensure compliance with the available resources and project objectives.

PROJECT SCOPE AND PLANNING

2.1. PROJECT SCOPE MANAGEMENT

2.1.1. PROJECT SCOPE STATEMENT (DESCRIPTION AND DEFINITION)

The project scope statement provides a structured definition of the activities, objectives, deliverables, and boundaries of the project. Its purpose is to clearly identify the work required for the development of the digital platform related to the historical memory of the Four Days of Naples, while reducing ambiguities and ensuring alignment among stakeholders throughout the project lifecycle.

Furthermore, the scope statement establishes the basis for project execution, monitoring, and acceptance, supporting effective coordination between managerial, legal, and technical activities.

1. Project Definition :

The project consists of the development of a web-based digital platform dedicated to the preservation and dissemination of historical and cultural content related to the Four Days of Naples. The platform is developed by Partenope Project Lab for the client Graf S.R.L. and aims to provide an accessible and interactive digital environment for citizens, students, researchers, and cultural institutions. The platform integrates multimedia materials such as photographs, videos, podcasts, articles, and narrative content in order to promote cultural engagement and facilitate the accessibility of historical knowledge through modern digital technologies.

2. Project Objectives :

The project objectives can be considered SMART:

- **SPECIFIC:** the platform must provide users with access to historical multimedia content related to the Four Days of Naples through an organized and interactive digital environment.
- **MEASURABLE:** the platform functionalities will be validated through testing activities, user feedback, and milestone approvals during the development process.
- **ACHIEVABLE:** the software development team, coordinated by the Project Manager, will ensure the implementation of the required functionalities within the available budget and timeframe.

- **RELEVANT:** the project contributes to the preservation of cultural heritage, the dissemination of historical knowledge, and the enhancement of the promoting organization's digital value.
- **TIME-BOUND:** the project is expected to be completed within approximately seven months according to the planned milestone structure.

3. Deliverables :

Several deliverables are expected during the project lifecycle.

During the initial phase :

- Functional Analysis documents.
- Definition of customer journeys and platform structure.
- Initial design and planning activities.

During the development phase :

- Development of the web platform and multimedia management functionalities.
- Integration of photographs, videos, podcasts, and narrative content.

-Development of accessibility and user interaction functionalities.

-Testing release for user validation.

During the final phase :

- Final validated release of the platform.
- Correction of identified defects and system refinements.
- Deployment of the platform on the client infrastructure.
- Final approval by the client and project tutors.

4. Out of Scope:

The following activities are considered outside the project scope:

- Development of dedicated mobile applications.
- E-commerce or monetization functionalities.
- Hardware management and physical infrastructure supply.
- Long-term operational maintenance after project completion.
- Physical preservation or restoration of historical documents.

5. Constrains :

One of the main project constraints concerns the limited timeframe available for development and delivery activities. The project is expected to be completed within approximately seven months, requiring continuous monitoring of activities and milestones.

Another important constraint is represented by the limited budget available for the initiative, estimated at approximately €57,000. Financial resources must therefore be carefully managed throughout the project lifecycle.

In addition, organizational constraints are present due to the relatively small size of the development team and the simultaneous management of multiple projects by Partenope Project Lab. Part of the software development activities also depends on external developers, increasing coordination and integration complexity.

6. Assumptions:

- Availability of multimedia historical content for integration into the platform.
- Sufficient user interest in accessing historical and cultural digital content.
- Successful collaboration between internal and external development teams.
- Successful validation of the platform by the client and funding initiative tutors.
- Effective coordination between managerial, legal, and technical activities throughout the project lifecycle.

7. Review and Updates:

To ensure effective monitoring and alignment during project execution, regular meetings and progress reviews are organized among project stakeholders. Internal communication activities include periodic status updates, operational meetings, and milestone validation sessions.

The scope statement may be updated if significant changes occur regarding project requirements, constraints, assumptions, or stakeholder expectations during the execution phase.

2.1.2. PROJECT SCOPE VERIFICATION (ACCEPTANCE CRITERIA)

The project scope verification process ensures that the developed platform and its functionalities comply with the requirements defined during the planning phase and agreed upon with the client. Verification activities are conducted progressively throughout the project lifecycle through testing sessions, milestone reviews, and validation meetings.

Particular attention is dedicated to the correct functioning of the platform, the integration of multimedia content, and the overall accessibility and usability of the system. The platform must operate without critical defects and provide a stable and interactive user experience.

The final acceptance of the project is achieved after the successful completion of testing activities, the validation of the deliverables by the client, and the positive evaluation of the tutors responsible for supervising the project within the funding initiative.

2.1.3. PROJECT SCOPE CHANGE CONTROL (CRITERIA FOR ANALYSIS AND RESPONSE TO POTENTIAL CHANGES)

The project scope change control process is intended to manage potential modifications requested during the development lifecycle while maintaining control over project activities, deadlines, and resources.

If significant change requests are proposed after the development activities have already started, these modifications are generally treated as formal Change Requests and are not directly included within the original project scope. In such cases, the requested functionalities or modifications are postponed and potentially implemented after the delivery of the platform through a separate maintenance or evolution contract.

On the other hand, smaller change requests may be evaluated during the project execution phase. In these situations, the Project Manager and the software development team assess the impact of the requested modifications in terms of time, cost, technical feasibility, and overall project quality. If considered manageable and compatible with the project constraints, these minor adjustments may be integrated into the ongoing development process.

This approach allows the project team to reduce the risk of scope creep while preserving flexibility and ensuring alignment with stakeholder expectations throughout the project lifecycle.

2.2. PROJECT PLANNING

Project planning formed the basis for organising and coordinating the activities required to develop and deliver the platform. Although the project involved several interdependent technical and non-technical activities, the planning approach was relatively flexible and less formalised. Rather than relying on extensive project management documentation, planning functioned mainly as a practical tool for coordinating activities, resources, responsibilities, and stakeholder expectations.

The project included technical development, design activities, content preparation, testing, documentation, stakeholder communication, and funding-related requirements. These activities needed to be aligned to ensure that the platform could be developed as a coherent final solution.

This section presents the project planning process through key project management areas, including integration management, work breakdown structure, quality, time, cost, human resources, risk, procurement, stakeholders, and communication. These areas are used to analyse how the project was structured, coordinated, and managed during the project lifecycle.

2.2.1. PROJECT INTEGRATION MANAGEMENT (CRITICAL ISSUES IN THE DEFINITION/MANAGEMENT OF ACTIVITIES ALSO IN RELATION TO THE REQUIREMENTS REQUESTED BY THE DIFFERENT STAKEHOLDERS)

Project Integration Management was essential in ensuring that activities, resources, and stakeholders were coordinated to deliver a coherent and consistent project outcome. Project integration management concerns the processes and activities needed to identify, define, combine, unify, and coordinate different project activities and resources (Annarelli, 2026e). This was particularly relevant in this project, as the work depended on coordination between technical development, design, content preparation, testing, stakeholder communication, and external development resources. The project was structured into five main phases: planning and initiation, design and prototyping, core development and feature implementation, testing and quality assurance, and documentation, finalisation and delivery. Integration management was therefore important both for coordinating the technical work across these phases and for maintaining alignment between the project's technical execution and stakeholder expectations.

PROJECT ROLES AND COORDINATION

The project involved both internal and external resources. The main roles included the Project Manager, Junior Developers, a Mid-level Developer, an internal communication team, designers, and an externally hired Senior Developer. The Project Manager was responsible for overall coordination, planning, stakeholder communication, reporting, and alignment between objectives, timelines, and requirements. The development resources were responsible for implementing technical functionalities, while the designers and communication team contributed to graphic prototyping, content-related activities, and alignment with the cultural purpose of the platform.

The externally hired Senior Developer contributed to selected technical activities, including functional specifications, system architecture, technology selection, application logic, and externally developed components. However, this role also represented an external supplier relationship, not only a technical development role. This distinction is important because the project depended on the timing, quality, and integration of externally delivered work.

INTEGRATION OF TECHNICAL AND NON-TECHNICAL ACTIVITIES

Integration was especially important because the project combined both technical and non-technical work streams. The technical work included database functionalities, authentication and access features, streaming and multimedia support, AI-related functionalities, accessibility features, and integrations with external platforms and services. These components had to function together as one unified platform, meaning that delays or errors in one part of the system could affect other activities.

At the same time, the project required coordination with non-technical activities such as UX/UI design, content collection, documentation, stakeholder communication, and client feedback. This made integration management important not only for the software components, but also for ensuring that the platform supported the broader cultural and communicative objectives of the project.

COMMUNICATION AND CHANGE MANAGEMENT

Collaboration and coordination were supported through periodic meetings and practical communication tools such as Trello, email, Google Drive, and WhatsApp. Trello was used to coordinate activities and monitor progress, while meetings and direct communication helped align the different work streams. Google Drive supported document sharing and collaboration, and email was used for more formal communication.

Changes in requirements were handled through official change requests, which were implemented in later releases according to specific contractual agreements. This helped control scope changes and ensured that new requirements were managed without disrupting the current development cycle.

INTEGRATION CHALLENGES

A significant integration challenge was related to the external supplier. The issue concerned both development timing and limited communication from the external supplier, which made coordination and integration more difficult. Since externally developed components had to be aligned with the internal team's work, insufficient communication and delayed delivery created additional integration work.

This challenge was handled by increasing operational resources. More specifically, an internal developer, mainly supporting integration and quality-related tasks, was assigned to work alongside the external supplier. This can be interpreted as horizontal integration, as the internal developer's responsibilities were broadened to include closer coordination with the external supplier. It also included elements of vertical integration, since the developer became more involved in evaluating, adapting, and supporting the integration of externally delivered components (Annarelli, 2026e). This helped improve coordination and ensured that external work could be better aligned with the internal team's technical standards. The case shows that integration management in the project was not only about combining technical components, but also about managing dependencies between internal and external contributors.

OVERALL INTEGRATION APPROACH

Several activities, especially development, testing, supervision, and coordination, were carried out in parallel. This supported faster progress and iterative improvement, but it also increased the need for continuous alignment between components, roles, and stakeholders. The project therefore followed a practical and partly Agile or hybrid-inspired approach, characterised by parallel activities, ongoing coordination, and adjustments during the project lifecycle.

Overall, Project Integration Management was central to coordinating work across phases, roles, resources, and stakeholder expectations. The project demonstrates that when parts of a software project are outsourced, integration management must include not only technical coordination, but also active follow-up of external dependencies, communication, deliverable quality, and responsibility boundaries. The case also shows how integration challenges may require both horizontal and vertical adjustments in project roles, as team members may need to take on broader coordination tasks and greater involvement in decision-making or quality assessment during the project lifecycle.

2.2.1.1. WORK BREAKDOWN STRUCTURE

In this analysis, the Work Breakdown Structure (WBS) is used to organise the project into manageable parts and define the overall structure of the project scope. In project management theory, a WBS is commonly used to break down project work into smaller components and work packages that can support planning, responsibility allocation, budgeting, scheduling, and progress control (Annarelli, 2026a). Based on the available project data, the project can be divided into five main phases:

1. Planning and Initiation
2. Design and Prototyping
3. Core Development and Feature Implementation
4. Testing and Quality Assurance
5. Documentation, Finalisation and Delivery

These phases represent the main areas of project activity and include both technical and managerial work. The phases are distributed across different project roles. The Project Manager is primarily responsible for planning, coordination, reporting, and stakeholder communication. The Senior Developer contributes to requirements definition, system architecture, technology selection, application logic, supervision, and technical-business alignment. The Mid-level Developer focuses on supervision, integration support, and quality assurance, while Junior Developers are mainly responsible for implementation tasks during the development-related phases.

The reconstructed WBS also reflects logical dependencies between activities. Planning and requirements definition form the basis for design and development activities, while testing and supervision depend on implemented functionalities. Several activities are carried out in parallel, particularly development and testing, reflecting a flexible and iterative workflow. Within these five main phases, the project is further divided into more detailed work packages and tasks, which form the basis for execution, coordination, and deliverable management.

Table 2.1 – Work packages

WBS ID	Phase	Work Package	Responsible Role(s)
1.1	Planning and Initiation	Define project requirements	Project Manager
1.2	Planning and Initiation	Prepare resource and financial planning activities	Project Manager
1.3	Planning and Initiation	Define sprint and milestone planning	Project Manager
2.1	Design and Prototyping	Define system architecture and application logic	Senior Developer
2.2	Design and Prototyping	Develop and validate UX/UI prototypes	Project Manager, Development Team
2.3	Design and Prototyping	Collect and organise platform materials	Communication Team, Client, Project Manager
3.1	Core Development and Feature Implementation	Configure database functionalities	Junior Developers, Senior Developer
3.2	Core Development and Feature Implementation	Implement authentication and access functionalities	Junior Developers, Senior Developer
3.3	Core Development and Feature Implementation	Develop advanced search functionality	Junior Developers, Senior Developer
3.4	Core Development and Feature Implementation	Implement subscription functionalities	Junior Developers, Senior Developer
3.5	Core Development and Feature Implementation	Develop streaming and multimedia functionalities	Junior Developers, Senior Developer
3.6	Core Development and Feature Implementation	Define and implement AI-related functionalities	Junior Developers, Senior Developer
3.7	Core Development and Feature Implementation	Integrate external platforms and services	Junior Developers, Senior Developer
4.1	Testing and QA	Conduct functional testing and supervision	Senior Developer, Mid-level Developer
4.2	Testing and QA	Perform beta and advanced testing	Development Team
4.3	Testing and QA	Verify integrations and validate functionalities	Senior Developer, Mid-level Developer
4.4	Testing and QA	Resolve bugs and implement corrections	Development Team
5.1	Documentation, Finalisation and Delivery	Prepare project documentation and reporting	Project Manager, Senior Developer
5.2	Documentation, Finalisation and Delivery	Conduct final review and platform adjustments	Project Manager, Development Team
5.3	Documentation, Finalisation and Delivery	Deliver final platform	Project Manager, Development Team

Table 2: Work Breakdown Structure

In this reconstructed WBS, the Senior Developer role refers to an externally hired developer who was responsible for selected technical design and development activities.

2.2.1.2. DESCRIPTION OF WORK PACKAGES, TASKS AND DELIVERABLES

The work packages identified in the reconstructed WBS represent groups of related activities that contribute to specific project outputs and deliverables. The work packages were reconstructed based on the available timesheet data, the project timeline, and observed project activities. They provide an analytical overview of how the project work can be structured, rather than a formal structure necessarily used by the company during execution.

The first work package, *Define Project Requirements*, focused on identifying and documenting the functional and business requirements of the platform. This included the detailed definition of functional specifications, stakeholder coordination, and validation of early wireframes and prototypes. The main deliverables were functional specifications and initial planning documentation. These activities were coordinated by the Project Manager, with the Senior Developer contributing to the detailed functional specification work.

The second work package, *Prepare Resource and Financial Planning Activities*, involved planning project resources, organising project activities, and supporting milestone and sprint planning. The outputs included project planning documentation and resource allocation structures, supporting the overall coordination of the project. This work package was mainly connected to the Project Manager's responsibilities.

The *Design System Architecture and Application Logic* work package focused on defining the technical structure of the platform. This included logical and physical system architecture, technology selection, application logic, and software module design. The main deliverables were the technical structure of the platform and the foundation for later development activities. Based on the timesheet data, the Senior Developer played a central role in this work package. Since this resource was externally hired, this work package also created an important dependency between the internal team and the external technical contribution.

The *Develop and Validate UX/UI Prototypes* work package included validation of wireframes, alignment between UX/UI and development activities, and early prototype testing. This work package contributed to the platform's interface structure and user experience. These activities were mainly coordinated by the Project Manager, with support from the development team where technical alignment was needed.

The project also included *Collection and Organisation of Platform Materials* as an important content-related work package. This involved collecting, preparing, and organising the historical, cultural, and multimedia materials to be inserted into the platform. This work package was important because the project was not only a technical development initiative, but also a cultural dissemination project. The main deliverables were structured content and materials ready for integration into the digital platform.

The largest group of work packages belonged to the *Core Development and Feature Implementation phase*. These included database configuration, authentication and access functionalities, advanced search, subscription management, multimedia support, AI-related functionalities, accessibility features, translation functionalities, and integration with external platforms and services. These work packages were mainly implemented by Junior Developers, while the Senior Developer and Mid-level Developer contributed through technical direction, supervision, integration support, and quality control.

Within this phase, several work packages contributed to the platform's core functionality. *Configure Database Functionalities* focused on project-specific database components, while *Implement Authentication and Access Functionalities* focused on login systems, page permissions, and user access functionalities. *Develop Advanced Search Functionality* and *Implement Subscription Functionalities* supported user interaction and access to platform content. *Develop Streaming and Multimedia Functionalities* covered video streaming and multimedia-related support, while *Define and Implement AI-related Functionalities* focused on AI algorithms and related system features. Finally, *Integrate External Platforms and Services* ensured that external platforms and services could interact with the system. This integration work was particularly important because several components depended on correct interaction in order to function as part of the overall platform.

Testing and quality assurance activities formed another important work package group. These activities included functional testing, beta testing, advanced testing, supervision of implemented functionalities, bug fixing, and verification of integrations. The development team was responsible for testing and correcting functionalities, while the Mid-level Developer supported supervision, validation, quality control, and integration. The main deliverables were tested functionalities, validation feedback, corrected system components, and bug fixes.

The final work package focused on *Documentation, Finalisation and Delivery*. This included project reporting, alignment between technical and business documentation, final review of project deliverables, final platform adjustments, and delivery of the completed platform. The outputs included final project documentation, reviewed deliverables, and a finalised version of the platform.

Overall, the identified work packages indicate a practical and task-oriented approach to project execution, where responsibilities and deliverables are closely linked to the organisation of activities throughout the project lifecycle. The reconstructed WBS also shows a clear distinction between coordination, technical definition, content preparation, implementation, testing, bug fixing, documentation, and delivery responsibilities.

2.2.2. QUALITY MANAGEMENT PLAN

The Quality Management Plan describes how the project ensured that the developed platform met functional expectations and supported reliable system performance throughout the development process. Project Quality Management can be understood as a set of activities aimed at ensuring that a project meets the needs for which it was undertaken. It normally includes quality planning, quality assurance, and quality control,

which define relevant quality standards, evaluate whether these standards are being met, and support corrective actions when necessary (Annarelli, 2026b).

In this project, quality was mainly associated with the correct functioning of platform features, system integration, usability, accessibility, and overall platform stability. Rather than relying on highly formalised quality management procedures, the project appears to have followed a practical and iterative approach, where quality assurance was integrated into daily development, supervision, testing, and coordination activities.

QUALITY APPROACH

Quality-related activities were distributed across multiple phases of the project rather than being concentrated in a single final testing phase. Testing and supervision were carried out continuously alongside development, which suggests an iterative approach where functionalities were validated, adjusted, and refined throughout the project lifecycle. This approach supported faster progress and made it possible to identify issues during development, but it also required continuous coordination between project roles to ensure that corrections and changes were properly integrated.

The project structure indicates that quality control was closely connected to communication, supervision, and practical collaboration within the team. Alignment activities, reviews, prototype validation, beta testing, and integration-related verification functioned as mechanisms for identifying and correcting issues during development. This flexible approach supported adaptability, but it also meant that quality assurance depended heavily on informal coordination rather than formal quality metrics or documented performance indicators.

TESTING AND FEEDBACK

Several testing approaches were used throughout the project. Unit testing was primarily performed by the developers during implementation activities, while User Acceptance Testing (UAT) was carried out with the involvement of testers. In addition, monkey testing was performed by the client in order to identify unexpected issues and validate system behaviour from a user perspective. Prototype testing, beta testing, advanced testing, and integration-related verification were also part of the quality assurance process.

User feedback was collected primarily by the Project Manager, who gathered feedback from the client and communicated relevant issues to the responsible project team or technical reference person. For example, platform-related issues were redirected to the software team responsible for the affected functionality. This created a continuous feedback loop between the client, project coordination, and the development team, helping the project respond to issues and stakeholder expectations during development.

ROLES AND RESPONSIBILITIES

Responsibility for quality management was shared across several project roles. The development team had the main responsibility for implementing and validating platform functionalities, while the Mid-level Developer contributed through supervision, validation, quality control, and integration support. The Project Manager supported quality-related coordination through reporting, alignment activities, communication with stakeholders, and feedback collection from the client.

The externally hired Senior Developer contributed to selected technical and architecture-related activities, but this role also created a quality-related dependency because parts of the work were outsourced. As a result, the quality of externally delivered components became relevant for the overall quality of the platform and the internal team's ability to integrate the work efficiently.

QUALITY CHANGES

A concrete quality-related challenge occurred in connection with the externally hired Senior Developer. Although this role was expected to contribute technical expertise and reduce the workload of the internal team, the delivered components were delayed and did not meet the required quality level. As a result, the internal team had to spend additional time correcting, adapting, and integrating the externally developed work. This shows that quality management was not only an internal development issue, but also depended on the quality control of outsourced technical work.

The overlap between development and testing also suggests a trade-off between time and quality. Continuous testing enabled faster progress while maintaining ongoing validation of implemented functionalities, but it also required close coordination to avoid inconsistencies between components. The issue with the external supplier further illustrates this trade-off, as poor-quality delivery created additional rework and integration pressure for the internal team.

2.2.3. TIME MANAGEMENT PLAN

The Time Management Plan describes how project activities are organised, monitored, and coordinated throughout the project lifecycle in order to ensure progress within the planned timeframe. It focuses on scheduling activities, coordinating dependencies between tasks, and monitoring progress across different phases of the project.

The project spans approximately seven months, from January 2024 to July 2024. The project timeline is structured around the same main phases identified in the reconstructed Work Breakdown Structure: Planning and Initiation, Design and Prototyping, Core Development and Feature Implementation, Testing and Quality Assurance, and Documentation and Finalisation. These phases overlap significantly rather than following a strictly sequential structure, reflecting the flexible and iterative nature of the project.

The project can be divided into the following main phases and approximate durations:

Phase	Approximate duration	Main activities
Planning and Initiation	January – April	Requirements definition, resource planning, financial planning, sprint and milestone planning
Design and Prototyping	January – March	System architecture, UX/UI alignment, prototype development and prototype validation
Core Development and Feature Implementation	March/April – July	Database configuration, implementation of platform functionalities, external integrations, AI-related functionalities
Testing and Quality Assurance	April – July	Functional testing, beta testing, advanced testing, supervision, validation of implemented features
Documentation and Finalisation	May – July	Reporting, technical and business documentation, final review, content production and platform adjustments

Table 3: Identified Phases of The Project

The project schedule is organised using a Gantt chart, which provides an overview of project activities, durations, and overlaps between phases. The schedule indicates that several activities are performed in parallel. In particular, development, testing, supervision, and coordination activities overlap throughout a large part of the project lifecycle. This structure supports continuous progress and iterative refinement of functionalities.

Here is a recreation of the Gantt chart the team used to plan:

Task	Start date	End date	Duration	61.01	08.01	15.01	22.01	29.01	05.02	12.02	19.02	26.02	05.03	12.03	19.03	26.03	02.04	09.04	16.04	23.04	30.04	07.05	14.05	21.05	28.05	04.06	11.06	18.06	25.06	02.07	09.07	16.07	23.07
Hiring Project Resources	01.01.2024	30.04.2024	120																														
Financial and Revenue Planning	01.01.2024	30.04.2024	120																														
Functional Prototype Development	19.01.2024	31.03.2024	73																														
LIS Association Collaboration	22.01.2024	31.07.2024	192																														
Functional Prototype Testing	31.01.2024	31.03.2024	61																														
Research Team Activities Phase 1	01.02.2024	30.04.2024	90																														
Information Platform Acquisition	01.03.2024	30.06.2024	122																														
Platform Development	01.04.2024	31.07.2024	122																														
Beta Testing	01.04.2024	31.07.2024	122																														
Research Team Activities Phase 2	01.05.2024	30.06.2024	61																														
Advanced Testing	01.05.2024	31.07.2024	92																														
Multimedia Content Production	01.05.2024	31.07.2024	92																														

Figure 3: Reconstructed Gantt Chart

MILESTONES AND PROGRESS MONITORING

Several milestones can be identified from the project timeline and project activities. These include the definition of functional requirements, completion and validation of initial prototypes, the start of platform development, the beginning of beta and advanced testing, and the final review and adjustment of the platform before completion. Tasks appear to have been organised and monitored through ongoing planning activities, sprint planning, milestone updates, reporting, and project coordination. The repeated presence of alignment tasks and timeline updates in the timesheet data suggests that the schedule was continuously monitored and adjusted throughout the project lifecycle.

Progress monitoring was primarily coordinated by the Project Manager through meetings, reporting activities, stakeholder communication, sprint planning, and milestone tracking. The Mid-level Developer also contributed through supervision, implementation follow-up, and validation of functionalities. This indicates that time management was handled through a practical and continuous control process rather than through a strictly fixed schedule.

FLEXIBLE AND PARALLEL TIME MANAGEMENT

The project structure indicates a flexible and iterative approach to time management. Rather than separating development and testing into isolated phases, testing and supervision were integrated alongside implementation activities. This allowed issues to be identified and corrected during development, but it also required continuous coordination between planning, development, testing, documentation, and external integration activities.

Several activities were carried out in parallel in order to maintain progress across multiple work streams. This overlapping structure helped the project move forward despite time constraints, but it also increased the need for coordination to avoid delays, inconsistencies, and dependency-related problems.

DELAYS AND SCHEDULE ADJUSTMENT

The project experienced delays partly due to the involvement of the externally hired Senior Developer. Since the internal team was also working on other projects, selected parts of the platform were assigned to this external developer. However, the externally delivered work required correction and integration effort from the internal team, which reduced the intended time-saving effect of outsourcing and created additional pressure on the project schedule.

These delays were managed through parallelisation of activities and reallocation of resources toward higher-priority tasks. In particular, operational resources were increased by assigning an internal developer to work alongside the external supplier. Although these actions did not eliminate the delays entirely, they helped limit the overall impact on the project timeline.

In the end, the project was delivered only a few days after the planned date, and this did not create major consequences for the client. This was partly because the delivery deadline had some flexibility, as the client

needed the platform for an exhibition scheduled for the following month. Delays and schedule adjustments were also communicated transparently to the client throughout the project lifecycle.

2.2.4. COST MANAGEMENT PLAN

The Cost Management Plan describes how project costs are estimated, monitored, and controlled throughout the project lifecycle. In this project, cost management is particularly important because the available budget is limited and the project must be completed within a defined financial framework.

The preliminary budget for the project was estimated at approximately **€57,000**. Based on the available timesheet data and hourly rates, the evaluated project cost amounts to **€56,940**, leaving a remaining budget of only **€60**. This indicates that the project was planned and evaluated very close to the available budget limit.

The cost structure of the project is primarily effort-based. This means that the main cost driver is the number of hours worked by project resources, multiplied by their hourly rates. The cost calculation follows a simple formula:

Total cost = hours worked × hourly rate

The total workload recorded in the timesheets amounts to 2,064 hours. These hours include project management, development, supervision, testing-related activities, documentation, and coordination work. Since this is a software development project, personnel effort represents the most significant cost element, while material or hardware costs appear to be less relevant.

The available cost data can be summarised as follows:

Resource category	Total hours	Hourly rate	Total cost
Development resource	384	€35.00	€13,440
Development / supervision resource	144	€25.00	€3,600
Junior development resource	360	€21.25	€7,650
Junior development resource	336	€21.25	€7,140
Junior development resource	312	€21.25	€6,630
Project management resource	256	€35.00	€8,960
Project management resource	272	€35.00	€9,520
Total	2,064		€56,940

Table 4: Cost Overview

The table shows that the project budget is mainly allocated to human resources. Development-related activities represent a substantial share of the cost, as they include platform implementation, database configuration, integrations, feature development, and technical adjustments. Project management also represents a significant cost element, reflecting the need for coordination, stakeholder communication, sprint and milestone planning, documentation, and reporting.

The project costs can also be understood in relation to the main phases identified in the WBS. Planning and initiation costs are mainly related to project management, requirements definition, and financial planning. Design and prototyping costs are linked to system architecture, UX/UI validation, and prototype testing. The core development phase represents the most resource-intensive part of the project, as it includes implementation of the platform's main functionalities. Testing and quality assurance costs are linked to supervision, validation, beta testing, and correction of issues. Finally, documentation and finalisation costs are related to reporting, documentation, final review, and platform adjustments.

Cost estimation was based on expected work effort, previous project experience, and the availability of operational resources. This is consistent with the nature of a small software organization, where cost planning is often based on estimated hours rather than complex financial models. The use of timesheets provides a practical basis for monitoring actual effort and evaluating whether the project remains within the available budget.

The externally hired Senior Developer was intended to increase development capacity and support progress within the limited project period. However, because the delivered work required correction and integration by the internal team, the outsourcing arrangement created indirect cost pressure.

Cost monitoring appears to be connected to regular project control activities, including timesheet tracking, reporting, invoice monitoring, client payments, and milestone follow-up. Cost follow-up was carried out through project status meetings, known as SAL meetings, where issued invoices and client payments were monitored. The Project Manager plays an important role in coordinating these activities and ensuring that costs remain aligned with project progress and the financial framework.

The budget is also connected to the funding framework of public programs such as the TOCC initiative, which supports digital and cultural projects. This makes cost control and reporting especially important, as the project must comply with funding-related documentation and reporting requirements.

2.2.5. INTEGRATED HUMAN RESOURCES MANAGEMENT PLAN

The Integrated Human Resources Management Plan explains how the project team was organised, including the allocation of roles and responsibilities needed to support effective teamwork, coordination, and project execution. The project involved both internal and external human resources. The internal project team consisted of Project Managers, Junior Developers, and a Mid-level Developer, while the Senior Developer

was an externally hired resource. This distinction is important because the Senior Developer was not only a technical contributor, but also represented an external dependency in the project.

2.2.5.1. ORGANIZATIONAL FUNCTIONS INVOLVED IN THE PROJECT

The main organisational functions involved in the project were project management, technical development, supervision and quality assurance, and external technical support. These functions were distributed across several project roles and supported different parts of the project lifecycle.

The Project Managers were responsible for the project management function. Their work included overall coordination, planning, stakeholder communication, reporting, sprint and milestone planning, timeline updates, UX/UI prototype validation, documentation, and alignment between technical and business perspectives. This function was important for ensuring that project activities remained coordinated and aligned with stakeholder expectations.

The technical development function was mainly carried out by the Junior Developers, who were responsible for implementing the platform's functionalities. Their tasks included database configuration, login and access functionalities, user and mailing list management, advanced search, subscription features, streaming, social media integration, accessibility features, translation functionalities, AI-related components, and integration with external platforms.

The Mid-level Developer contributed through supervision, validation, quality control, and integration support. This function was important for ensuring that implemented features were technically aligned and could be integrated into the overall platform.

The Senior Developer contributed to selected technical and architecture-related activities, including functional specifications, system architecture, technology selection, application logic, and specific software components. Since this role was externally hired, it also required coordination between the internal team and the outsourced technical resource, particularly to ensure that externally delivered work could be integrated with the rest of the platform.

2.2.5.2. PROJECT TEAM STRUCTURE

The project team had a relatively simple but collaborative structure. Strategic and coordination-related responsibilities were handled mainly by the Project Managers, while technical implementation was carried out by the Junior Developers. The Mid-level Developer supported the team through supervision, validation, quality control, and integration support. The externally hired Senior Developer contributed technical expertise to selected parts of the project, but was not part of the internal team in the same way as the other development resources.

This structure allowed the project to distribute work according to expertise and available capacity. However, the involvement of an external Senior Developer also meant that the team structure included both internal and external resources. As a result, coordination was not only needed within the internal project team, but also between the internal team and the external developer. This made communication, follow-up, and clear responsibility boundaries important for maintaining progress and ensuring that the externally delivered work could be integrated into the platform.

At a high level, the project's human resources can be summarised as follows:

Role	Main responsibilities
Project Managers	Planning, coordination, stakeholder communication, timeline updates, reporting, documentation, and alignment between technical and business perspectives
Senior Developer	Selected technical development tasks, architecture-related contributions, technology selection, application logic, and externally delivered components
Mid-level Developer	Supervision, validation, quality control, integration support
Junior Developers	Implementation of platform functionalities, technical tasks, and development of core system components

Table 5: Project Roles and Responsibilities

2.2.5.3. ASSIGNMENT OF RESPONSIBILITIES

Responsibilities in the project were assigned according to role, expertise, and project needs. The Project Managers were responsible for coordinating the overall project and maintaining alignment between objectives, timelines, stakeholders, and deliverables. They also acted as the main link between the client, stakeholders, and the development team.

The *Junior Developers* were assigned implementation tasks related to the platform's core functionalities. These responsibilities included both front-end and back-end related development activities, depending on the specific functionality being implemented. Their work formed the main technical foundation of the platform.

The *Mid-level Developer* was assigned responsibilities related to supervision, validation, quality assurance, and integration support. This role helped ensure that the implemented functionalities were technically correct, aligned with project requirements, and suitable for integration into the final platform.

The externally hired Senior Developer was assigned selected technical responsibilities, particularly related to architecture, technology choices, application logic, and specific development components. However, because this role was external, the assignment of responsibilities also created a supplier-related dependency. The internal team depended on the external developer's ability to deliver work that matched the project's requirements, technical standards, and integration needs.

A RACI matrix is a responsibility assignment tool used to clarify how roles are involved in different project activities. The letters stand for Responsible, Accountable, Consulted, and Informed. "Responsible" refers to the role performing the work, while "Accountable" refers to the role with final ownership of the activity or deliverable. "Consulted" refers to roles that provide input or expertise, while "Informed" refers to roles that should be kept updated about progress or decisions (Annarelli, 2026e).

The following RACI matrix has been reconstructed based on the available project information, as no formal RACI matrix appears to have been used during the project. It should therefore be understood as an analytical reconstruction of how responsibilities were distributed across the main project roles.

Project phase	Project Manager	Senior Developer	Mid-level Developer	Junior Developers	Communication Team	Client
Planning and Initiation	A/R	C	I	I	I	C
Design and Prototyping	A	R	C	C	C	C
Core Development and Feature Implementation	A	C	C	R	I	I
Testing and Quality Assurance	A	C	R	R	I	C
Documentation and Finalisation	A/R	C	C	R	I	C
Stakeholder Communication	A/R	I	I	I	C	C
Funding and Reporting Compliance	A/R	I	I	I	I	C

Table 6: Reconstructed RACI matrix

The matrix is based on the five main project phases identified in the Work Breakdown Structure: Planning and Initiation, Design and Prototyping, Core Development and Feature Implementation, Testing and Quality Assurance, and Documentation and Finalisation. In addition to these phases, the matrix includes two

cross-cutting management activities: Stakeholder Communication and Funding and Reporting Compliance. These activities are included because they were not limited to one specific project phase, but supported coordination, reporting, and alignment throughout the project lifecycle.

Overall, the reconstructed RACI matrix supports the analysis of the project's practical and role-based approach to human resource management. It shows how responsibilities were distributed according to functional needs, technical expertise, and available capacity, while also highlighting the importance of clear responsibility boundaries between internal and external project contributors

2.2.6. RISK MANAGEMENT PLAN

The Risk Management Plan describes how potential project risks were identified, assessed, and managed in order to reduce their impact on project objectives. In project management theory, risk can be understood as an uncertain event or situation that may affect one or more project objectives, including time, cost, and quality. Risk management may follow either a proactive approach, where risks are identified, evaluated, and prevented before they materialise, or a reactive approach, where problems are addressed through correction, crisis management, and compensation of negative effects after they occur (Annarelli, 2026c).

In this project, there was no highly formalised risk evaluation process or separate risk register used during execution. Instead, risk management appears to have been handled in a practical and iterative way through project coordination, sprint planning, milestone follow-up, stakeholder communication, and continuous monitoring of project activities. This suggests that the project relied more on reactive and adaptive risk management than on a fully proactive risk management framework.

PROJECT RISK

Based on the available project information, the main risks in the project were related to time constraints, limited internal capacity, dependency on external technical resources, integration complexity, quality issues, and stakeholder expectations. These risks were particularly relevant because the project involved several interdependent technical components, including database configuration, authentication, streaming, AI-related functionalities, accessibility features, multilingual support, and integrations with external platforms and services.

One of the most significant risks was dependency on an externally hired Senior Developer. Since the internal team was also involved in other projects, parts of the platform were assigned to this external resource in order to increase development capacity. However, this created a dependency on the external developer's ability to deliver work on time and at the required quality level. This risk materialised when the externally developed components required additional correction and adaptation before they could be integrated into

the platform. As a result, the internal team had to spend extra time on rework and integration, which increased pressure on both the schedule and the development process.

Another important risk was schedule pressure. The project was carried out within a limited timeframe of approximately six months, while several activities were performed in parallel. Development, testing, supervision, coordination, and documentation overlapped throughout the project lifecycle. This flexible structure supported progress, but it also increased the risk of delays, inconsistencies, and reduced control over dependencies between tasks. The delay and quality issues related to the external developer further increased this risk.

Quality risk was also relevant throughout the project. Since the platform included several complex and interconnected features, insufficient testing or poor-quality implementation could affect the stability and usability of the final solution. Quality risks were managed through continuous testing, supervision, validation of implemented functionalities, beta testing, user acceptance testing, and feedback collection from the client. However, the issue with the external developer shows that clearer acceptance criteria and stronger quality checks for outsourced work could have reduced the amount of rework required later in the project.

A further risk concerned stakeholder expectations. The project involved both technical and non-technical stakeholders, and the platform had to meet expectations related to functionality, usability, accessibility, and progress. If stakeholder requirements were unclear or changed during the project, this could affect scope, priorities, and delivery expectations. This risk was managed through regular communication, reporting, prototype validation, meetings, and continuous alignment between the Project Manager, client, and development team.

Another relevant risk was connected to the project's funding framework. Since the budget was aligned with public funding through the TOCC initiative, the project had to comply with specific financial constraints, documentation requirements, and reporting obligations. If the project failed to meet these requirements, this could create administrative challenges, delays in reporting, or potential issues related to funding compliance. This risk was therefore connected not to the funding itself, but to the need to manage the project in accordance with the conditions attached to the funding scheme.

The project team appears to have managed risks mainly through reactive and iterative control actions rather than through a formal risk management framework. Relevant mitigation actions included reallocating internal resources, carrying out activities in parallel, integrating testing into the development process, maintaining frequent communication with stakeholders, and using supervision to identify technical issues early. These actions helped reduce the impact of risks, although they did not eliminate all consequences, particularly those related to the external development dependency.

The main project risks can be summarised as follows:

Risk	Possible impact	Mitigation or control action
Dependency on externally hired Senior Developer	Delays, rework, integration problems, and reduced control over deliverable quality	Follow-up, internal correction, additional integration work, and stronger coordination between internal and external resources
Limited internal capacity	Slower development progress and increased workload on internal team members	Outsourcing selected work packages and reallocating internal resources when needed
Schedule pressure	Delays and reduced flexibility in handling unexpected issues	Parallelisation of activities, sprint planning, milestone tracking, and continuous progress monitoring
Integration complexity	Technical inconsistencies between components and additional coordination needs	Continuous integration support, supervision, validation, and testing
Quality issues	Reduced platform stability, usability problems, and need for rework	Unit testing, UAT, beta testing, supervision, client feedback, and quality control
Stakeholder expectation misalignment	Scope changes, unclear priorities, or dissatisfaction with project progress	Regular meetings, reporting, prototype validation, and continuous communication
Funding and reporting compliance risk	Administrative delays, documentation issues, or potential non-compliance with funding requirements	Regular cost monitoring, timesheet tracking, invoice monitoring, reporting, and alignment with the funding framework

Table 7: Project Risk Register

Overall, the project did not have a formalised risk evaluation process with documented probability and impact scoring. However, several risks were identified and managed throughout the project in a practical way. The project team responded to risks through coordination, communication, testing, supervision, and resource adjustments. The experience with the externally hired Senior Developer particularly highlights the importance of identifying supplier-related risks early and defining clearer quality requirements, acceptance criteria, and follow-up mechanisms when parts of the development work are outsourced.

2.2.7. PROCUREMENT MANAGEMENT PLAN

The Procurement Management Plan describes how external resources and services were used in the project, and how these affected project execution. In project management theory, procurement management has both an operational and strategic role, as external suppliers can influence project performance in terms of time, cost, quality, and integration (Annarelli, 2026d).

Based on the available project information, there is limited evidence of a formal procurement management process. The clearest procurement-related element was the use of an externally hired Senior Developer, who

was brought into the project to increase development capacity while the internal team was also working on other projects. However, this external resource also created a dependency, as the project relied on the supplier's ability to deliver work on time, at the required quality level, and in a form that could be integrated into the platform.

This dependency became a procurement-related challenge when the externally delivered work required additional correction and adaptation by the internal team. Such issues are consistent with common procurement challenges, including delivery delays, lower-than-expected quality, and the need for closer monitoring of supplier work (Annarelli, 2026d). As a result, the expected benefit of outsourcing was partly reduced, while the need for coordination, quality control, and integration work increased.

Overall, procurement management in the project appears to have been relatively informal and practical. The case shows that even limited outsourcing can create significant project dependencies, especially when externally delivered work is closely connected to the platform's core functionality. For similar projects, clearer expectations and stronger follow-up of external deliverables could reduce the risk of delays, rework, and integration problems.

2.2.8. STAKEHOLDER MANAGEMENT PLAN

The Stakeholder Management Plan describes how the main stakeholders in the project were identified, analysed, and involved throughout the project lifecycle. In project management theory, stakeholders are individuals, groups, or organisations that may affect, be affected by, or perceive themselves to be affected by a project decision, activity, or outcome. Stakeholder management therefore involves identifying relevant stakeholders, understanding their interests and influence, and determining how they should be involved or informed during the project (Arnone, 2025).

KEY STAKEHOLDERS

The key stakeholders in the project included the client, Graf S.R.L., the project organisation Partenope Project Lab, the Project Manager, the internal development team, the externally hired Senior Developer, the TOCC funding framework, and the final users of the platform.

Graf S.R.L. was the client and one of the most important external stakeholders. The client's main interest was to receive a functional digital platform that could support the dissemination of historical and cultural content related to the Four Days of Naples.

Partenope Project Lab was responsible for developing the platform and managing the project execution. As the project organisation, Partenope Project Lab, had an interest in delivering the solution within the agreed scope, timeframe, budget, and quality expectations.

The *Project Manager* had a central coordination role and acted as the main link between the client, stakeholders, and the development team. The internal development team was responsible for implementing, testing, and integrating the platform functionalities.

The externally hired *Senior Developer* was also a relevant stakeholder, as this role contributed to selected technical activities and represented an external dependency. The TOCC funding framework influenced the project through financial constraints, deadlines, and reporting requirements. Finally, the *final users*, including citizens, students, and researchers, were important because the platform was designed to meet their needs for accessible and engaging cultural content.

STAKEHOLDER INTEREST AND INFLUENCE

The stakeholders had different levels of power and interest depending on their role in the project. The client, the project organisation, and the Project Manager had both high influence and high interest, as they were directly involved in defining, managing, and delivering the project. The TOCC funding framework also had high influence because it affected the financial conditions, reporting requirements, and deadlines, although its direct involvement in daily project activities was more limited. In contrast, the internal development team had high interest because the project directly affected their workload, priorities, and technical tasks, but their decision-making power was lower than that of the client, Project Manager, and project organisation.

The final users, including citizens, students, and researchers, had high interest but low direct power. Their needs influenced the platform's focus on usability, accessibility, multimedia storytelling, and structured access to historical content, even though they were not directly involved in daily project decisions. The externally hired Senior Developer had medium influence, as the quality and timing of his deliverables affected the project's technical progress and integration work.

STAKEHOLDER INVOLVEMENT AND COMMUNICATION

Stakeholders were involved in different ways depending on their role and influence in the project. The client was involved through feedback, validation, and approval of project direction and deliverables, while the Project Manager coordinated communication between the client and the development team. The developers were mainly involved through implementation, testing, and technical alignment activities. Communication was handled through meetings, reporting, project updates, and ongoing dialogue, suggesting a practical and continuous communication approach rather than a highly formalised stakeholder communication system.

The Project Manager was mainly responsible for managing stakeholder communication. This role was important for collecting feedback, communicating requirements, monitoring progress, and ensuring that stakeholder expectations were translated into concrete project activities.

STAKEHOLDER RELATED CHALLENGES

A key stakeholder-related challenge was the need to balance different expectations and constraints. The client and funding framework expected a culturally valuable and accessible digital platform, while the development team had to manage technical complexity, limited resources, and time constraints. In addition, the externally hired Senior Developer created a coordination challenge because externally delivered work had to be aligned with the internal team's technical standards and integration needs. This made follow-up, communication, and clear responsibility boundaries important.

Overall, stakeholder management in the project was based on continuous communication, coordination, feedback collection, and alignment between the client, project management, and technical resources.

2.2.9. COMMUNICATION MANAGEMENT PLAN

The Communication Management Plan describes how information was shared within the project team and with relevant stakeholders. In project management, communication can be understood as the exchange of information between project stakeholders to ensure alignment, shared understanding, and informed decision-making. This includes verbal, written, digital, and other forms of interaction (Arnone, 2026).

Communication was particularly important in this project because it involved several internal and external contributors, including the Project Manager, Junior Developers, a Mid-level Developer, the communication team, designers, the client, and an externally hired Senior Developer. Since the project combined technical development, content-related work, design activities, testing, client feedback, and externally delivered components, clear communication was necessary to coordinate tasks, manage expectations, follow up dependencies, and address issues during the project lifecycle.

INTERNAL COMMUNICATION AND TOOLS

Communication within the project team was handled through a combination of periodic meetings and digital communication tools. Trello was used to coordinate activities, monitor progress, and keep track of tasks, while email was used for more formal communication and updates. Google Drive supported document sharing and collaboration, particularly for project documentation and materials, while WhatsApp was used for faster and more direct communication between project participants. Together, these tools supported both formal and informal communication needs and helped the team coordinate work across different roles and activities.

MEETINGS AND INFORMATION SHARING

Project meetings were used to coordinate activities, discuss progress, identify issues, and align priorities. The available information suggests that meetings were held periodically rather than according to a highly formalised meeting schedule. In addition to coordination meetings, project status meetings, known as SAL meetings, were used to monitor project progress, issued invoices, and client payments. This means that

communication was not only related to technical progress, but also to administrative and financial follow-up.

The information shared during the project included task progress, requirements, change requests, development issues, testing feedback, bug reports, financial updates, and decisions related to priorities and releases. Changes in requirements were handled through official change requests and implemented in later releases according to specific contractual agreements. This helped ensure that changes were communicated and managed in a controlled way, rather than being added informally during ongoing development.

STAKEHOLDER COMMUNICATION

The Software Project Manager had the main responsibility for communication and acted as the link between the client, the development team, and other project stakeholders. This role was responsible for collecting client feedback, communicating requirements, following up on progress, and forwarding issues to the relevant team or technical reference person. For example, if a problem concerned the platform, the Project Manager reported it to the software reference person for the project.

Communication with the client was handled through updates, feedback collection, reporting, and transparent communication about delays. When delays occurred, they were communicated openly to the client. This was important for managing expectations and maintaining alignment between the client and the project organisation throughout the project.

COMMUNICATION CHALLENGES

A communication-related challenge occurred in relation to the external supplier. The project experienced limited communication from the external supplier, which contributed to coordination difficulties and affected the timing of development work. This issue was addressed by increasing operational support, specifically by assigning an internal developer to work alongside the external supplier. This helped improve coordination and supported the integration of externally delivered work into the platform.

CHAPTER 3.

PROJECT EXECUTION AND CONTROL

3.1. SCOPE MANAGEMENT AND CONTROL

Scope Management and Control ensured that the "Napoli Liberazione" project delivered the intended digital platform features — including multimedia storytelling, AI functionalities, and accessibility tools — without unauthorized deviations from the defined requirements. As established in the scope statement, the project focused exclusively on the digital platform and its content management capabilities, excluding any physical installations or offline services. Given the flexible and partly Agile-inspired approach adopted throughout the project, scope control was not managed through rigid, formalised documentation, but rather through continuous alignment, iterative feedback, and practical coordination between the Project Manager, the development team, and the client.

3.1.1. CONTROL ACTIONS AND MEASURES

Scope control actions were integrated into the daily development and coordination activities rather than being concentrated in dedicated review gates. The primary tool for monitoring task progress and verifying that activities remained within the defined scope was Trello, which allowed the project team to track the implementation of core functionalities — such as database configuration, authentication, streaming, and external integrations — against the work described in the work packages.

The Software Project Manager played a central role in scope verification, acting as the main link between the client (Graf S.R.L.) and the development team. This role involved collecting client feedback, validating UX/UI prototypes, and communicating requirements to ensure that the technical execution remained aligned with both the cultural objectives and the functional specifications agreed upon at the start of the project. Periodic SAL meetings provided additional control points where project progress, open issues, and priorities could be reviewed and aligned with client expectations.

Continuous supervision by the Mid-level Developer further supported scope control by verifying that implemented features matched the functional specifications before integration into the overall platform. Together, these mechanisms created a continuous feedback loop that kept the project scope under active monitoring throughout the development lifecycle.

3.1.2. SCOPE CHANGE REQUEST, SCOPE CREEP AND CORRECTIVE ACTIONS

To prevent scope creep — the uncontrolled expansion of project requirements — the project relied on a structured approach to managing modifications. Changes requested by stakeholders were handled through official change requests and, rather than being introduced immediately into the ongoing development cycle, were scheduled for implementation in later releases according to specific contractual agreements. This mechanism successfully protected the core development phase from continuous interruptions and preserved the team's focus on delivering the originally planned functionalities.

A significant issue that indirectly threatened the project scope arose in connection with the externally hired Senior Developer. The externally delivered software components were delayed and did not meet the required quality standards, forcing the internal team to dedicate additional time and effort to correction and adaptation work. If left unmanaged, this rework burden could have created pressure to reduce the platform's scope — for example, by cutting planned features — in order to meet the final delivery deadline.

However, the corrective action taken prioritised scope preservation over schedule relief. Rather than reducing the planned deliverables, operational resources were increased by assigning an internal developer to work directly alongside the external supplier, supporting both the correction of external components and their integration into the platform. This measure allowed the project to maintain its original scope and deliver all planned functionalities, despite the technical and coordination challenges encountered. The case highlights how effective scope control in this project was not only about managing formal change requests, but also about protecting the defined scope from the indirect pressures created by external dependencies.

3.2. QUALITY MANAGEMENT AND CONTROL

Quality Management and Control focused on ensuring that the implemented functionalities, system integrations, and accessibility features met the required technical and user expectations throughout the project lifecycle. Rather than relying on rigid, highly formalised quality metrics, the project adopted a practical and iterative control process where quality assurance was embedded directly into the daily development and supervision cycle, consistent with the hybrid and Agile-inspired approach described in the Quality Management Plan.

3.2.1. CONTROL ACTIONS AND MEASURES

To maintain continuous alignment with project requirements, multiple testing methodologies were applied concurrently with development activities. Unit testing was performed continuously by Junior Developers during the implementation phase, providing an early and ongoing check on the stability of individual

software components. User Acceptance Testing (UAT) was subsequently carried out to validate usability, accessibility features, and system integration from a functional standpoint, ensuring that implemented features matched both the technical specifications and user expectations. In addition, monkey testing was conducted directly by the client to deliberately interact with the platform in unexpected ways, helping to surface edge cases and unanticipated system behaviours that structured testing might not have captured.

The Mid-level Developer played a crucial role in quality control throughout this process, providing continuous supervision, code validation, and integration support. This ensured that individual components were technically aligned before being merged into the overall platform architecture. Simultaneously, the Project Manager facilitated an ongoing feedback loop by collecting defect reports and client observations and redirecting them promptly to the appropriate technical reference within the development team. This coordination between testing, supervision, and client feedback created a continuous quality monitoring mechanism that operated in parallel with development rather than as a separate, end-of-cycle activity.

3.2.2. PROBLEMS DETECTED AND CORRECTIVE ACTIONS

The most critical quality-related problem encountered during the execution phase was linked to the externally hired Senior Developer. As anticipated in the Risk Management Plan, the software components delivered by this external resource were not only delayed but also failed to meet the project's technical standards, lacking the necessary alignment for seamless integration into the main platform. This deficiency introduced a significant volume of unplanned technical debt that had to be absorbed by the internal team.

As a corrective action, internal developers were required to allocate substantial additional time to review, debug, adapt, and properly integrate the flawed external components. This situation created a direct tension between schedule pressure and quality requirements: reducing the rework effort would have risked releasing unstable functionalities, while fully addressing the quality issues required time that had not been budgeted for this purpose.

The project's iterative approach — where testing and development activities overlapped throughout the lifecycle — proved important in this context, as it allowed the quality issues to be identified relatively early rather than only at final delivery. However, the episode also exposed a structural weakness in the quality management approach: the absence of clearly defined acceptance criteria and formal quality checks for outsourced components meant that substandard work could enter the integration pipeline before being detected. This represents one of the key corrective insights to be drawn from the project experience, and will be further discussed in the Lessons Learned section.

3.3. ACTIVITY PROGRESS MANAGEMENT AND CONTROL

Activity Progress Management and Control focused on monitoring the execution of project activities against the planned seven-month timeframe, from January to July 2024. Because the project followed a flexible and highly parallelised structure — where development, testing, supervision, and coordination activities heavily overlapped — continuous schedule monitoring was essential to prevent bottlenecks, manage task dependencies, and ensure that progress across multiple work streams remained aligned.

3.3.1. CONTROL ACTIONS AND MEASURES

To track progress effectively, the project relied on continuous iterative planning rather than a rigid, static schedule. The overall timeline was initially structured using a Gantt chart (see Section 2.2.3), providing an overview of phase durations, overlaps, and key milestones, as described in the Time Management Plan. However, day-to-day progress was actively managed through sprint planning sessions, milestone tracking, and periodic status meetings known as SAL (Stato Avanzamento Lavori) meetings. These meetings served as structured control points where activity progress, open issues, issued invoices, and client payments were reviewed, ensuring that both technical and administrative dimensions of the project remained on track.

The Project Manager held primary responsibility for monitoring the timeline, updating milestone statuses, and coordinating task prioritisation through Trello. This was complemented by the daily use of Timesheets by all team members, allowing the Project Manager to track the actual effort (man-hours) spent against the planned schedule. This tool provided a shared and continuously updated view of ongoing activities, making it possible to identify emerging delays or dependency conflicts before they escalated. Concurrently, the Mid-level Developer provided continuous supervision over the technical implementation, ensuring that development tasks advanced at the expected pace and that parallel testing activities remained properly synchronised with the ongoing development cycles.

3.3.2. PROBLEMS DETECTED AND CORRECTIVE ACTIONS

The most significant threat to the project schedule arose, once again, from the involvement of the externally hired Senior Developer. This external resource had been procured specifically to increase development capacity and support the delivery of selected platform components within the constrained timeframe. However, the delayed submission of external components — combined with the additional rework required to bring them to an acceptable quality level — completely negated the anticipated time savings and introduced unplanned pressure across the entire project schedule.

Two main corrective actions were implemented in response, utilizing standard schedule compression techniques. First, the team applied Crashing—a technique aimed at shortening the project schedule by adding resources to critical path activities, trading cost for time. In practice, an internal developer was assigned to work directly alongside the external supplier, accelerating the correction and integration process and reducing the dependency on the external resource's pace of delivery. Second, the team leveraged a Fast Tracking approach—a schedule compression technique where activities or phases that would normally be

performed sequentially are instead executed in parallel. By applying this, the team reorganised its workflow to increase the parallel execution of unaffected activities, helping to partially recover lost time without reducing the planned scope of the platform

Despite these pressures, the Project Manager maintained full transparency with the client, Graf S.R.L., communicating schedule adjustments promptly and ensuring that stakeholder expectations remained aligned with the realistic delivery timeline. This approach reflected a broader principle applied throughout the project: that proactive communication and flexible resource management were more effective responses to schedule variance than attempting to conceal delays or reduce deliverable scope. The experience ultimately confirmed that, in projects with limited internal capacity and external dependencies, schedule control must include not only progress monitoring but also active follow-up on the timing and quality of outsourced contributions.

3.4. COST MANAGEMENT AND CONTROL

Cost Management and Control represented one of the most critical aspects of project execution, given the tight budget of €57,000 and the strict financial reporting requirements associated with the TOCC public funding framework. As established in the Cost Management Plan, the project's cost structure was entirely effort-based, meaning that expenditures were directly driven by the number of hours worked by each resource multiplied by their respective hourly rates. This structure made timesheet accuracy and resource allocation discipline the two most important levers for maintaining financial control throughout the project lifecycle.

3.4.1. CONTROL ACTIONS AND MEASURES

To ensure the project remained within its financial boundaries, cost monitoring was tightly integrated with schedule and resource management activities. The primary control mechanism was continuous timesheet tracking, which allowed the Project Manager to evaluate actual effort expenditure against the planned workload in near real time. By maintaining an up-to-date record of hours worked across all resource categories — from Junior Developers to project management roles — the team could detect early signs of budget pressure and adjust resource prioritisation accordingly.

Financial progress was formally reviewed during periodic SAL (Stato Avanzamento Lavori) meetings, where issued invoices, client payments, and cumulative timesheet records were assessed together to verify alignment between project progress and budget consumption. This approach ensured that cost control was not treated in isolation but remained consistently connected to both technical progress and the reporting obligations of the TOCC funding framework. Given the effort-based cost model described in the Cost

Management Plan — where the total evaluated cost of €56,940 left a remaining contingency of only €60 — this level of monitoring was not optional but essential to avoid any risk of non-compliance.

3.4.2. PROBLEMS DETECTED AND CORRECTIVE ACTIONS

The most significant cost-related challenge encountered during the execution phase was the indirect financial pressure generated by the externally hired Senior Developer. The outsourcing arrangement had originally been designed to optimise resource allocation within the constrained budget, by shifting selected technical responsibilities to an external contributor without exceeding the planned cost envelope. However, because the externally delivered components required substantial correction and adaptation by the internal team, internal resources were forced to absorb unplanned rework hours that had not been budgeted for this purpose.

Given the negligible contingency margin of €60, even a modest volume of unplanned internal effort risked pushing the total project cost beyond the €57,000 threshold, which would have created compliance issues with the TOCC funding requirements. As a corrective action, the Project Manager enforced rigorous timesheet oversight and systematically re-prioritised the remaining development activities to absorb the additional effort within the existing budget. By carefully reallocating internal resources and optimising the sequencing of lower-priority tasks, the team succeeded in integrating the necessary rework without requesting supplementary funds, thereby maintaining full financial compliance with the funding framework.

The episode highlights a broader lesson relevant to effort-based cost management —a financial model where project expenditures are directly driven by the number of hours worked by each resource multiplied by their respective hourly rates. Because time is strictly equivalent to cost in this approach, when contingency margins are minimal, any unplanned dependency — particularly one involving outsourced work of uncertain quality — can rapidly transform a financial plan that appears controlled into one that is structurally fragile. Proactive cost monitoring, combined with early identification of supplier-related risks, is therefore essential in projects operating under similarly constrained financial conditions.

3.5. RISK MANAGEMENT AND CONTROL

Risk management in the Napoli Liberazione project was not a separate process. It ran through the same mechanisms the team used to track schedule and cost: SAL meetings, timesheet reviews, client feedback, and the Mid-level developers day-to-day technical supervision. There was no standalone risk register with probability scores refreshed at fixed intervals. What existed instead was situational awareness built into routine coordination. That worked, until it did not.

3.5.1. CONTROL ACTIONS AND MEASURES

The SAL meeting structure served as the primary control point. At each session, the Project Manager assessed technical progress, resource consumption, and client alignment together rather than separately. Timesheet tracking did double duty: it monitored costs, and it flagged pressure building in any work stream. Unexpected effort in one area often meant rework, integration friction, or a dependency slipping.

This approach partially reflects the risk control process defined in project risk management literature, where continuous monitoring should draw on performance data to update impact assessments and probability estimates across the lifecycle (Annarelli, 2026c). In this project, timesheet data and SAL meeting outputs function as a performance data source, though without a formal risk register update cycle tied to those reviews.

The Mid-level Developer handled continuous quality monitoring across implemented components. Problems surfaced during development rather than at final delivery. Given how tightly the platform's components were interconnected, this made a real difference.

The client relationship also functioned as a risk signal. The Project Manager maintained regular contact with Graf S.R.L. and used milestone validations and prototype reviews to detect expectation gaps before they became delivery disputes.

The contract between the client and partenope entailed different articles that shaped the monitoring context in concrete ways. The contract is therefore used in this section to describe the monitoring procedures of the project. Two contract clauses shaped the monitoring context in concrete ways. [Article 7](#) tied payment to five specific deliverables: the contract signature, the functional analysis, the test release, the pre-final release, and the validated final release. Delays in delivery meant delays in incoming payment, which created direct cash flow pressure on Partenope Project Lab. [Article 6](#) required the team to communicate significant delays caused by technical reasons promptly, and gave the client the right to withdraw after 180 days of delay. These were not abstract incentives. They meant that undiscovered schedule drift had a specific contractual consequence.

3.5.2. IDENTIFIED ISSUES AND UPDATE OF RISK REGISTER

The most consequential risk during execution corresponded to what the team identified in the Business Case as R02: dependency on the externally hired Senior Developer. Although no formal risk register existed in the actual project, this risk had been recognised and categorised as high likelihood and high impact during the academic analysis of the project. Engaging external developers always carries inherent risk, since the commissioning organization has limited direct control over delivery quality and timing. In this project, that general risk materialized concretely: the external developer delivered components that were both late and technically below standard, putting simultaneous pressure on schedule, quality, and cost at the same time.

The team responded by crashing and fast tracking: an internal developer was assigned to work alongside the external supplier, and parallel workstreams were reorganised to recover lost time. The response contained the damage. It did not eliminate it.

R01 and R03, the schedule pressure risk and the general resource strain risk, did not materialise independently. They were amplified by R02. The outsourcing arrangement that was supposed to relieve schedule pressure instead consumed more of it.

R04, quality inconsistency, also tied back to the external developer. The team had identified this risk and framed its mitigation around supervision and coordination. What the mitigation did not include was contractually embedded quality gates for third-party deliverables. That absence meant substandard components entered the integration pipeline before anyone caught them. The supervision strategy was right in principle, the checkpoint structure was not built to enforce it.

The funding and reporting risk did not materialise. The Project Manager enforced strict timesheet oversight and reprioritised activities to absorb rework costs within the original budget. The final evaluated cost came to 56,940 euros, leaving 60 euros of contingency. Financial compliance with the TOCC funding framework held.

Stakeholder expectations were managed without serious incident. Consistent transparency with the client about delays kept the relationship stable through the disruption.

The clearest lesson from this project is not about risk registers, but about outsourced work. R02 was identified as a high risk before the project started, and the mitigation relied mainly on supervision. However, supervision is not a substitute for defined acceptance criteria and formal checkpoints for third-party deliverables. The next project using external developers should therefore consider having quality gates written into the supplier arrangement, rather than monitored informally alongside it.

3.6. PROCUREMENT MANAGEMENT AND CONTROL

Procurement management in the Napoli Liberazione project appears limited in the available documentation, but it was still consequential for project execution. Based on the available data, the clearest procurement-related decision was the engagement of an externally hired Senior Developer to supplement internal development capacity. Given the scope of the project, it is reasonable to assume that some procurement-related procedures may also have been involved, such as defining the external resource's role, agreeing on deliverables, managing contractual expectations, and following up on the quality and timing of the work. However, these procedures are not clearly documented in the available material.

The analysis should therefore be understood as focusing on the procurement elements that can be identified from the project data, rather than as a complete assessment of the company's procurement practices. What can be observed is that the external development contribution created an important project dependency.

When the externally delivered work required correction and adaptation by the internal team, procurement became relevant not only as a sourcing decision, but also as a matter of supplier follow-up, quality control, and integration management.

3.6.1. CONTROL ACTIONS AND MEASURES

The primary control mechanism was the milestone-based payment structure embedded in the contract between Partenope Project Lab and Graf S.R.L. **Article 7** in the contract tied disbursements to five deliverable events: the contract signature, the functional analysis, the test release, the pre-final release, and the validated final release. This created indirect financial accountability. Delays or quality failures in externally delivered components would postpone the internal teams own milestones and their payment. It was a meaningful incentive, but payment gates were spaced across the full project arc, which made them too infrequent to catch technical deficiencies within individual development phases before those phases were complete.

The Project Manager served as the main interface with the external resource, following up through the SAL meeting structure and timesheet tracking. SAL meetings covered technical progress, resource consumption, and outstanding issues, providing a practical coordination rhythm. Coordination, though, is not quality verification. The SAL structure was effective at monitoring overall progress and budget consumption, it was not configured to evaluate the technical adequacy of specific components before they entered the integration pipeline.

3.6.2. PROBLEMS DETECTED AND CORRECTIVE ACTIONS

The core procurement failure identified was the absence of defined acceptance criteria for externally delivered work. When the Senior Developer submitted components for integration, there was no documented quality specification, no formal inspection step before handover, and no agreed remediation process for non-compliance. Deficient components entered the integration pipeline before their quality level was fully understood. That compressed the detection-to-correction cycle and displaced internal effort allocated to other tasks.

Article 3 of the Graf S.R.L. contract required Partenope to perform commissioned activities with professional diligence appropriate to the nature of the work. Article 5 explicitly permitted use of third-party subcontractors, but placed full responsibility for their performance on Partenope Project Lab, with associated costs borne by Partenope. The contract created a clear accountability structure. The project's internal procurement controls were not built to match the oversight that structure implied. Contractual responsibility was accepted, the corresponding quality verification mechanisms were not put in place.

The corrective action, described in Section 3.3, involved assigning an internal developer to work alongside the external supplier to support correction, adaptation, and integration of the flawed components. This

resolved an acute problem, but the internal developer was effectively performing the acceptance review that should have been built into the procurement arrangement from the start. His role shifted from development support to quality remediation, absorbing unplanned hours and generating the indirect cost pressure documented in Section 3.4.

For future projects with a similar structure, the case suggests that externally delivered work should be supported by clearer verification mechanisms during execution. This could include formal checkpoints and defined technical criteria for determining when external contributions are ready for integration. Such measures would strengthen the internal team's ability to monitor supplier performance, identify quality issues earlier, and reduce the risk of rework in later project phases.

3.7. PROJECT TEAM AND HUMAN RESOURCES MANAGEMENT

The team structure from the planning phase carried over into execution without major changes: Project Managers coordinated, a Mid-level Developer supervised and handled quality assurance, and three Junior Developers did most of the implementation work. What the project had not accounted for was an external procurement failure that would redirect internal capacity partway through the project. The platform's technical scope was already large, covering database configuration, authentication systems, multimedia streaming, AI features, accessibility, multilingual support, and external integrations. Managing that scope while absorbing work from an outside hire that failed to deliver put real pressure on the team for most of the project lifecycle.

3.7.1. CONTROL ACTIONS AND MEASURES

The main visibility tool was continuous timesheet tracking. Every team member logged hours against specific activities, and the Project Manager reviewed actual effort expenditure in near real time. Over the full lifecycle, the team recorded 2,064 hours across development, supervision, project management, and coordination. When a work package started consuming more hours than planned, that data gave the Project Manager concrete grounds for adjusting priorities in the next sprint, without waiting for a formal replanning event.

Sprint planning turned that visibility into decisions. Instead of locking in a fixed weekly task list, the team used sprint cycles to rebalance priorities against actual progress. Schedule pressure in one area could be addressed within the current sprint by shifting effort elsewhere. This suited the project's structure: development, testing, and supervision overlapped for much of the lifecycle, and the sprint rhythm managed that overlap without creating unnecessary process overhead.

The Mid-level Developer played a control function that went beyond the formal supervision role. This person sat closest to both the Project Managers coordination view and the Junior Developers daily work. By

catching integration issues early and surfacing problems before they became schedule delays, the Mid-level Developer reduced the risk of effort being quietly absorbed by undetected technical difficulties. Given the platform's interconnected architecture, where a problem in one component could propagate into others quickly, that early-warning function was not incidental.

3.7.2. PROBLEMS DETECTED AND CORRECTIVE ACTIONS

The biggest disruption came from the external Senior Developers failure to deliver components at the required quality level and on schedule. Junior Developers originally allocated to building core platform features were pulled into unplanned rework, adaptation, and integration correction instead. This was a substantial shift. The three Junior Developers together accounted for 1,008 hours in the timesheet records, redirecting a portion of that capacity toward correction work directly disrupted the planned effort distribution across the development phases.

The impact was uneven. Work packages that depended on the externally delivered components could not advance until those components were corrected and verified. Junior Developers working on interconnected functionality were either blocked or forced to work around incomplete deliverables. The Mid-level Developers supervisory load increased during this period, since the additional integration and verification work required more intensive oversight than a standard development cycle would have demanded.

The corrective action, detailed in Section 3.3, was to assign an internal developer to work alongside the external supplier and accelerate the correction process. This was a deliberate trade: capacity that could have gone toward new development was redirected to absorb what the external arrangement had failed to produce. The reallocation recovered enough schedule flexibility to meet the final delivery requirements, but it did not restore the original effort distribution. The internal team carried the cost of the external failure in absorbed rework hours. The 60 euros remaining in the budget at project completion is the clearest measure of how precisely that additional internal effort had to be managed against a total financial envelope of 57,000 euros.

3.8. STAKEHOLDER MANAGEMENT AND CONTROL

Stakeholder management during execution was not a parallel process running alongside schedule and cost control. It was built into the same coordination rhythm. The Project Manager served as the primary interface between Graf S.R.L. and the internal development team, maintaining active communication throughout the lifecycle. This served two purposes: keeping the client informed, and protecting the relationship during the disruption caused by the external developers' delivery failures. Where the project absorbed stakeholder pressure without losing trust, transparency did the work.

3.8.1. CONTROL ACTIONS AND MEASURES

The primary mechanism for client engagement was the SAL meeting structure. These progress review sessions, documented as early as 22 January 2024 in the first Stato Avanzamento Lavori report, gave Graf S.R.L. a structured, recurring window into project status. The first SAL reported completed activities, including the selection of accessibility and inclusivity guidelines and the drafting of functional requirements and test plans, alongside pending activities and a detailed list of open questions requiring client input.

The format communicated controlled progress and invited the client to resolve ambiguities before a milestone, rather than discovering them at one. The open questions logged in the SAL document covered content timelines, collaboration with research and creative teams, account registration features, and QR code placements. All required client decisions before development could move forward. By surfacing these dependencies explicitly, the Project Manager kept Graf S.R.L. engaged as an active participant rather than a passive recipient.

The contract reinforced this structure financially. Article 7 tied the total project fee of 57,000 euros to five specific deliverable events, from contract signature through the validated final release. Article 9 defined the corresponding payment schedule, with the largest single installment, 20,000 euros, linked to product delivery in a test environment for user validation. This architecture gave Graf S.R.L. direct financial visibility into each development phase. Delays in delivery translated immediately into delays in disbursement, giving the client concrete contractual reasons to monitor progress closely and respond promptly to milestone submissions. Articles 10 and 11 formalized the clients role further, establishing a verification and acceptance process for intermediate and final releases, including the production of a signed test report documenting identified defects. The client was not simply approving outputs, it was performing quality validation as a contractual obligation.

Monkey testing, carried out directly by Graf S.R.L., extended this validation role into the development cycle itself. By interacting with the platform in unstructured and unexpected ways, the client contributed to defect detection in ways no structured testing script would replicate. This deepened the clients sense of ownership over the platform and gave the Project Manager an additional feedback channel for identifying and prioritizing corrections.

3.8.2. CHANGES IN STAKEHOLDER INFLUENCE/INTEREST DETECTED AND UPDATE OF THE STAKEHOLDER REGISTER/MAP

The stakeholder map established in the planning phase remained broadly stable during execution. Several shifts in effective influence and engagement intensity, however, are visible in the project record.

Graf S.R.L. maintained consistently high engagement throughout. The milestone-based payment structure gave the client an ongoing financial incentive to stay informed, and the SAL meeting format provided the

mechanism to act on that interest. The first SAL document illustrates this clearly: the client was not being updated on progress, it was being asked to make decisions that shaped the direction of development. Graf S.R.L.'s practical influence over technical and content decisions increased during execution as open questions accumulated and required client resolution before work could continue.

The TOCC funding framework operated as a stakeholder with delayed but escalating influence. Its requirements shaped the financial reporting structure and documentation standards from the start, but the pressure intensified as reporting deadlines approached in the later phases. The strict timesheet oversight and the care taken to keep total evaluated costs within the 57,000 euro ceiling, described in Section 3.4, were driven in part by TOCCs documentation and reporting obligations. The funding framework's practical influence on internal decision-making was higher in the final months than the planning-phase analysis implied.

The externally hired Senior Developer began the project as a medium-influence contributor, responsible for selected technical components the internal team depended on. That influence did not persist. As the quality and timing of the external contributions proved inadequate, the internal team progressively absorbed the affected work packages, reducing the dependency. By the point a dedicated internal developer was assigned to work alongside the external supplier, the external developer's effective influence had diminished substantially. The supplier remained formally present, but the internal team controlled the integration and correction process directly.

End users, including citizens, students, and researchers, maintained high interest throughout but exercised no direct influence on execution decisions. Their needs came through indirectly: via the accessibility requirements embedded in the platform specifications, the LIS collaboration documented in the SAL progress report, and client feedback from monkey testing and prototype validation sessions. For the execution phase, this was adequate. The tradeoff is that the platform's usability assumptions were filtered through the client's interpretation rather than tested with the intended audience directly.

3.9. COMMUNICATION MANAGEMENT AND CONTROL

The communication framework from Section 2.2.9 held up well in practice. Trello gave all internal contributors a shared, live view of task progress across parallel work streams, without requiring dedicated sync meetings. WhatsApp and direct messaging handled fast, informal coordination, issues surfaced and got resolved quickly at the working level. SAL meetings served as the main structured touchpoint, covering technical progress, resource consumption, and outstanding issues in a single session. Google Drive supported document sharing throughout the lifecycle. None of these tools or formats created notable problems during execution.

Article 23 of the contract between Partenope Project Lab and Graf S.R.L. required all contractually relevant communications to be submitted in writing to specified addresses. This placed a formal layer on top of the

day-to-day tools, particularly for anything touching deliverables, delays, or contract conditions. The SAL meeting cycle and written reports provided the paper trail Article 23 required.

Mapping the project's communication activities onto the standard framework of frequency, format, and recipients reveals a clear pattern. SAL meetings provided low-frequency, high-formality structured reporting to Graf S.R.L. WhatsApp and Trello provided high-frequency, low-formality coordination among the internal team. The gap in the framework was the absence of any structured communication channel toward the external Senior Developer, which meant that when his responsiveness declined, no formal escalation path existed.

3.9.1. PROBLEMS DETECTED AND CORRECTIVE ACTIONS

Two communication failures emerged during execution. Each had a distinct cause and called for a different response.

The first involved the external Senior Developer. His limited responsiveness left the internal team without visibility into the status and quality of his work. Trello and WhatsApp could not compensate for that gap because both tools depend on active participation from all parties. Problems accumulated in the external work stream and went undetected until the affected components reached the integration pipeline. The failure was not simply one of delivery, it was a communication failure that delayed the team's ability to identify and respond to a quality problem. The fix was structural. An internal developer was assigned to work directly alongside the external supplier, converting what had been a remote, asynchronous handoff into an active collaboration. That restored the information flow the suppliers limited communication had cut off, and gave the internal team real-time visibility before integration was attempted.

The second failure concerned the relationship with Graf S.R.L. during the delays caused by the external developer. The Project Manager communicated the delays early and explicitly, framing them alongside the corrective actions already underway. This was not a discretionary choice. Article 6 of the contract required Partenope Project Lab to notify the client promptly whenever delays arose from technical causes, and it gave Graf S.R.L. the right to withdraw if a notified delay exceeded 180 days. Early disclosure was a contractual obligation. It also protected the relationship. As documented in the first SAL report of 22 January 2024, the communication structured with Graf S.R.L. was built to surface open issues and pending decisions explicitly, treating the client as an active participant rather than a passive recipient of status updates. Keeping that format through the disruption kept Graf S.R.L. informed, reduced the risk of the client being caught off guard, and satisfied the notification requirements under Article 6 without leaving the project in undisclosed breach.

CHAPTER 4.

PROJECT CLOSURE AND CRITICAL EVALUATION

4.1. PROJECT CLOSURE

The Project Closure phase marked the formal completion of the "Napoli Liberazione" platform and the final transfer of all deliverables to Graf S.R.L. Following the final UAT cycle, the client provided formal sign-off through a signed test report certifying that all identified defects had been resolved, in accordance with the contract Articles. The project team then executed the formal handover of the platform and all technical and business documentation. Financially, The Project Manager closed the execution respecting the TOCC budget threshold, ensuring full compliance with the funding initiative's reporting obligations. Internal and external resources were formally released, and the platform was deployed on the client's infrastructure. Long-term maintenance remains out-of-scope and is expected to be addressed through a separate arrangement if required.

4.2. ANALYSIS OF CRITICAL SUCCESS FACTORS AND RESULTS (PERFORMANCE AND BENEFITS OBTAINED)

The project was delivered with its full planned scope intact. All features defined in the Project Charter — multimedia storytelling, content management, AI functionalities, user authentication, subscriptions, multilingual support, accessibility tools, and external platform integrations — were implemented and validated. Preserving full scope under schedule pressure represents one of the most significant outcomes of the project.

The final delivery milestone was met in July 2024 within the planned seven-month duration. Schedule recovery required deliberate application of crashing and fast-tracking techniques, with an internal developer assigned to correct and integrate the external supplier's components, while workstreams were reorganised to recover lost time. These actions were communicated transparently to Graf S.R.L. under Article 6 of the contract.

At €56,940 against a €57,000 budget (99.9% utilisation), financial control was maintained despite unplanned rework, through rigorous timesheet oversight and activity re-prioritisation. Given the TOCC framework's requirements, this was as much a compliance result as a financial one.

The platform was delivered without critical defects, passing both UAT and client-led monkey testing. The parallel testing and development approach throughout the lifecycle allowed defects to be identified and resolved during development, significantly reducing their impact on schedule and cost.

BENEFITS REALISATION

The project delivered value across cultural, social, and strategic dimensions. The platform digitises and preserves the historical memory of the Four Days of Naples, making it accessible to students, researchers, and citizens through a multilingual and accessibility-compliant environment. For Graf S.R.L., the platform represents a durable digital asset supporting future cultural initiatives. For Partenope Project Lab, the project reinforced its demonstrated capability in managing hybrid software development under strict budget and regulatory constraints.

CRITICAL SUCCESS FACTORS

Three factors were decisive. First, the adoption of a hybrid management framework combining milestone planning with agile execution and continuous testing, which allowed the team to absorb the external supplier's underperformance without scope or quality failure.

Second, proactive resource mitigation: the rapid reallocation of an internal developer to work alongside the external supplier was the key operational intervention that recovered the schedule.

Third, stakeholder transparency: consistent communication with Graf S.R.L. regarding delays and corrective actions maintained trust and kept client expectations aligned throughout execution.

4.3. LESSONS LEARNED

The project generated five key lessons for future initiatives.

The most significant concerns supplier management: relying on external expertise without formally defined acceptance criteria, structured inspection gates, and a contractual remediation process creates disproportionate risk. Supervision alone is not a substitute for embedded quality controls, particularly when the contract assigns full accountability for subcontractors, as under Article 5.

A related lesson concerns internal resource flexibility. The ability to rapidly redeploy an internal developer was the only factor that prevented a critical schedule failure, highlighting the value of T-shaped competencies within small teams managing concurrent projects. Internal versatility is a more reliable buffer against supplier disruptions than contingency reserves alone.

The third lesson concerns the hybrid execution framework. Overlapping development and testing throughout the lifecycle — rather than concentrating quality checks at the end — surfaced defects early,

when correction costs were manageable. For future projects, this approach should be formally adopted rather than treated as an informal cultural characteristic.

The fourth lesson concerns contingency planning. Closing with only €60 remaining illustrates how structurally fragile a zero-contingency plan is. The financial outcome reflected the team's responsiveness, not the plan's resilience. Future publicly funded projects should include a risk-adjusted contingency of three to five percent, which the TOCC framework's requirements provide legitimate grounds to justify.

The fifth lesson concerns documentation discipline. Early investment in rigorous timesheet tracking and administrative reporting reduced pressure during closure and ensured the accuracy of final financial reporting required by the funding framework. Documentation is not merely a bureaucratic obligation but a compliance instrument.

4.4. CRITICAL EVALUATION OF THE PROJECT

In critically evaluating the “Napoli Liberazione” project, it is evident that its ultimate success was achieved through execution resilience rather than flawless initial planning. The project serves as a compelling case study on the relationship between structural project vulnerabilities and adaptive management capabilities: it succeeded not because every risk was anticipated and eliminated, but because the team's response capacity proved sufficient to contain the consequences of the risks that materialised.

From a methodological perspective, the decision to implement a hybrid project management framework — combining traditional milestone planning with agile execution and continuous, overlapping testing cycles — was the most effective strategic choice made during the project lifecycle. This adaptability allowed the internal team to absorb the severe impact of the external supplier's underperformance without triggering a scope or quality failure. Had a strict, sequential waterfall approach been adopted, the poor quality and delays of the external deliverables would have likely caused an unrecoverable schedule baseline rupture during the final integration phase, with consequences that no amount of reactive resource reallocation could have fully repaired.

However, a critical evaluation must also identify the structural vulnerabilities that created the conditions for the disruption in the first place. Three areas warrant direct assessment.

The first concerns procurement and supplier management. The decision to engage an external Senior Developer was rational given the internal team's capacity constraints, but the project did not build the control structure required to manage an outsourced technical dependency of this importance. The absence of formal acceptance criteria, pre-integration inspection steps, and contractually defined remediation procedures meant that substandard components entered the integration pipeline before their quality level was fully understood. This was not an isolated operational difficulty: it was a systematic gap that affected

schedule, quality, and cost simultaneously. The contractual responsibility accepted under Article 5 was not matched by the operational mechanisms needed to exercise it. A more structured supplier management approach, even a proportionate and lightweight one suited to the organisation's size, would have reduced the exposure materially.

The second area concerns contingency design. A budget margin of €60 on a €57,000 project leaves virtually no room for unplanned events. The project maintained financial compliance, but only because of intensive monitoring and precise resource reallocation under sustained pressure. This outcome cannot be treated as evidence that the financial plan was adequate: it reflects the team's capacity to respond, not the resilience of the plan itself. For future projects of this type, a minimum contingency reserve, even three to five percent of total budget, should be negotiated as part of the initial funding application. The TOCC framework's requirements provide legitimate grounds for presenting a risk-adjusted estimate rather than a zero-contingency plan.

The third area concerns the degree of process formalisation. The project was managed effectively through a practical and pragmatic approach, and in many respects this suited the organisation's size and operational culture better than a heavily formalised framework would have. Nevertheless, the absence of a documented risk register with explicit probability and impact scoring, the informality of the procurement management process, and the reliance on supervision rather than structured checkpoints for outsourced work all represent areas where a modest increase in formalisation would have provided meaningful protection. The project's experience with the external developer illustrates precisely the failure mode that structured processes are designed to prevent.

In conclusion, the "Napoli Liberazione" project can be classified as a definitive success in terms of product delivery and stakeholder satisfaction. A fully functional, culturally significant, and accessible digital platform preserving the memory of the Four Days of Naples was delivered on time and within budget, with the complete technical scope intact and the client relationship stable at closure. This outcome fulfilled the primary objectives established in the first chapter: to develop an innovative, inclusive, and technically advanced platform capable of maximising accessibility, engagement, and cultural impact. From a project management standpoint, the experience highlights a fundamental principle: while robust procurement controls and proactive contingency planning are essential preconditions for reliable project execution, it is ultimately the team's agility, transparency, and capacity for rapid problem-solving that determine whether a project succeeds when plans encounter reality.

APPENDIX

REFERENCES

- Annarelli, A. (2026, February 26). Principles of Project Management [Lecture Slides].
- Annarelli, A. (2026, March 11). Project Time Management [Lecture Slides].
- Annarelli, A. (2026, March 12). Project Cost Management [Lecture Slides].
- Annarelli, A. (2026, March 17). Project Quality Management [Lecture Slides].
- Annarelli, A. (2026, March 18). Project Risk Management [Lecture Slides].
- Annarelli, A. (2026, March 25). Project Procurement Management [Lecture Slides].
- Annarelli, A. (2026, March 30). Integration—RACI matrix [Lecture Slides].
- Annarelli, A. (2026, March 30). Project Integration Management [Lecture Slides].
- Arnone, V. (2026, April 1). Stakeholder Engagement [Lecture Slides].
- Arnone, V. (2026, March 19). Project Communication Management [Lecture Slides].
- Partenope Project Lab Team. *Personal interview with the project development team.*